SIEMENS

MAGNETOM

MR

Functional Description 1

System Information

Vision

Vision plus

Register 1
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English
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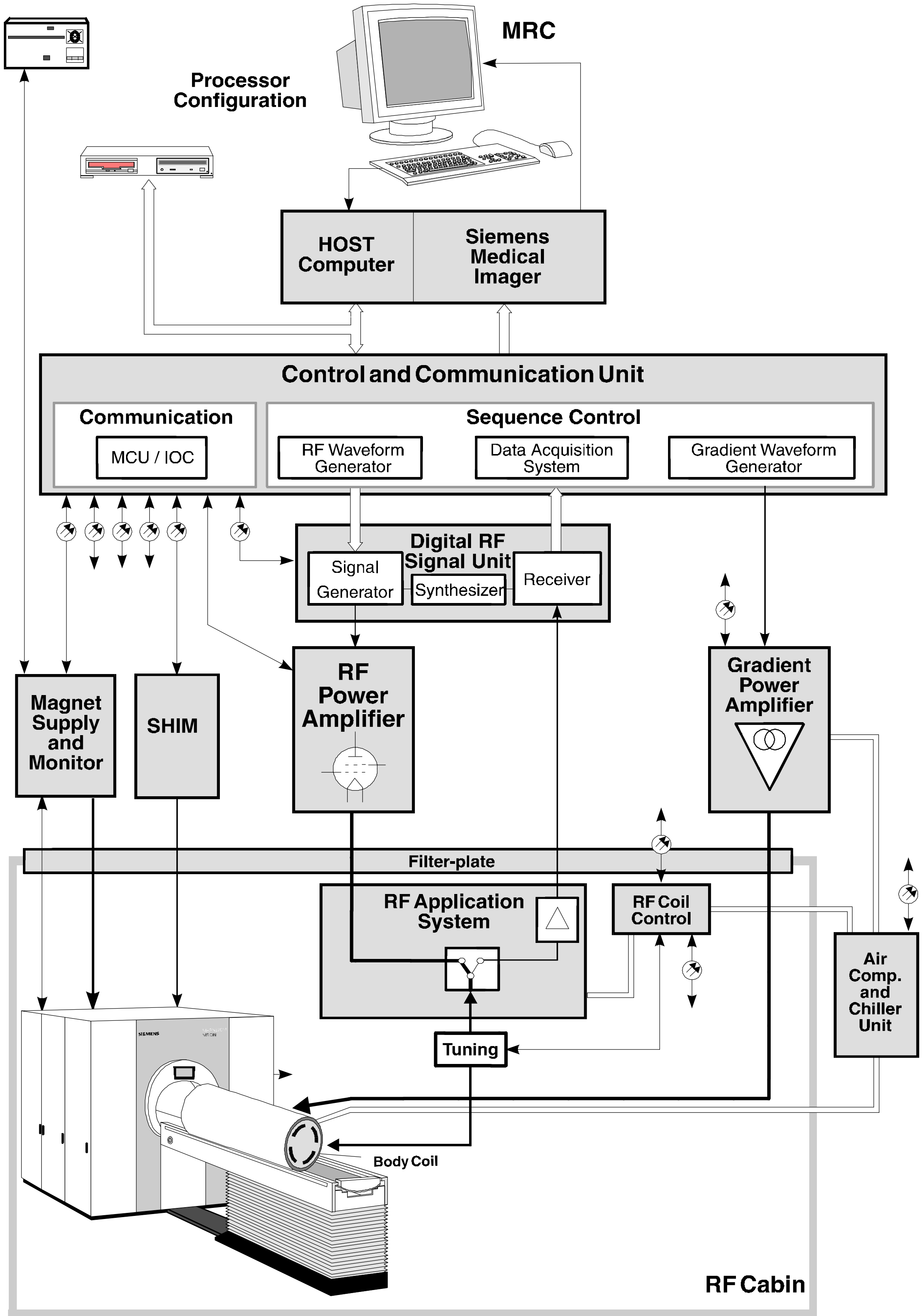
Vision System
Components
Overview



MRC
Processor Configuration
HOST Computer
Siemens Medical Imager
Control and Communication Unit
Communication
MCU / IOC
Sequence Control
RF Waveform Generator
Data Acquisition System
Gradient Waveform Generator
Digital RF Signal Unit
Signal Generator
Synthesizer
Receiver
RF Power Amplifier
Gradient Power Amplifier
Magnet Supply and Monitor
SHIM
Filter-plate
RF Application System
RF Coil Control
Air Comp. and Chiller Unit
Tuning
Body Coil
RF Cabin

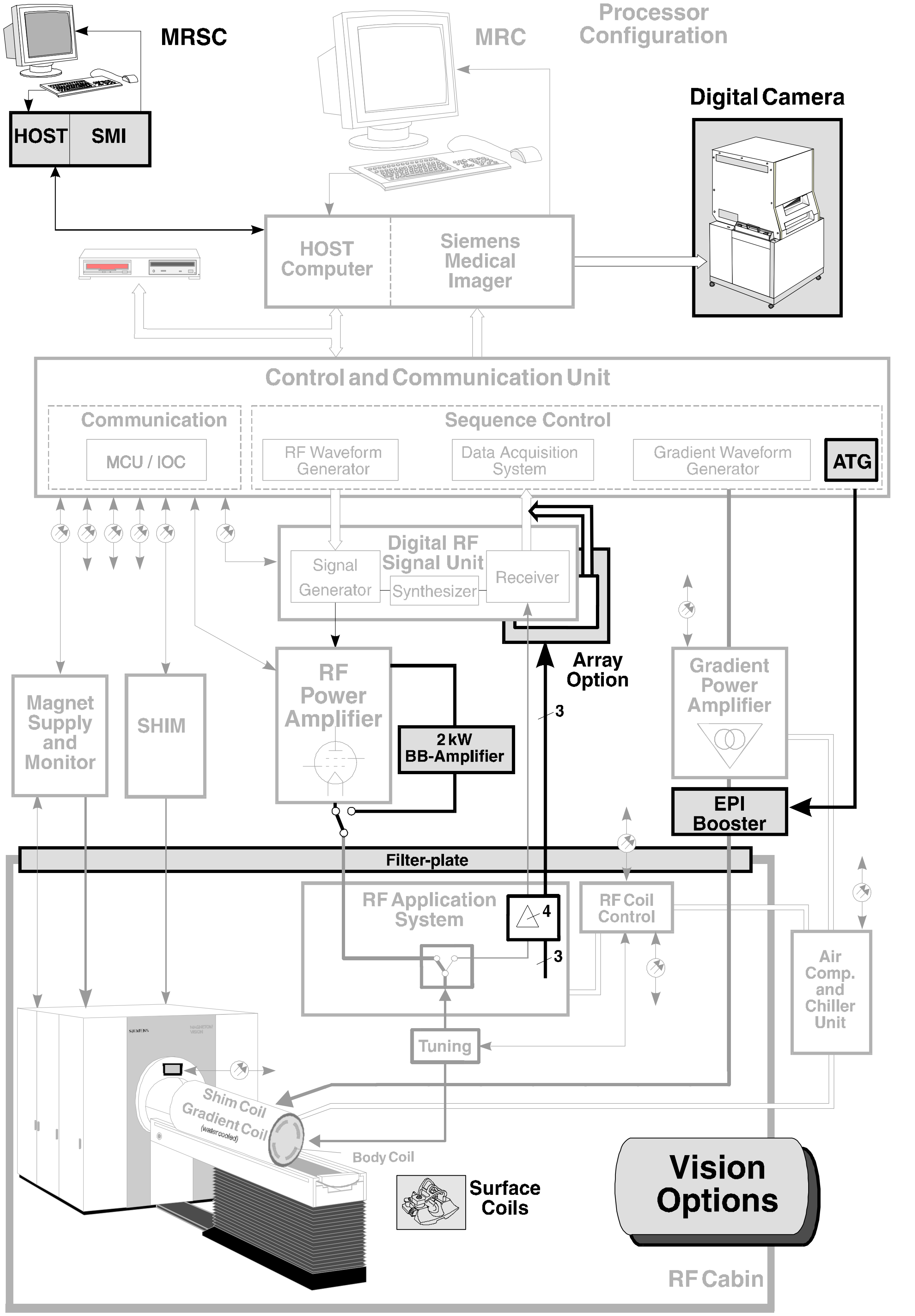
Introduction

The main components which comprise the Vision MR system are :

- Processor Configuration (**PC**)
  - **HOST** computer
  - MR Console (**MRC**)
  - Siemens Medical Imager (**SMI-5**)
- Control and Communication System (**CCU**)
- RF System
  - Digital RF Signal Unit (signal generator and receiver)
  - RF Power Amplifier (**RFPA**)
  - RF Application System (**RFAS**)
  - RF Coil Control electronics (coil tuning, dynamic de-tuning)
- Gradient System
  - Gradient Power Amplifier (**GPA**)
- Patient Table (**PT**)
- Magnet System
  - Magnet OR35 (active shielded and screened)
  - Magnet Power Supply (**MPS**)
  - Magnet Monitoring System (**MMS**)
  - Shim

Air Compressor and Chiller Unit (**ACCU**)

These components will be looked at here with the intent to convey a general overview of their functionality and interconnections while pointing out major advancements and advantages.



MRSC
HOST
SMI
MRC
Processor Configuration
Digital Camera
HOST Computer
Siemens Medical Imager
Control and Communication Unit
Communication
MCU / IOC
Sequence Control
RF Waveform Generator
Data Acquisition System
Gradient Waveform Generator
ATG
Digital RF Signal Unit
Signal Generator
Synthesizer
Receiver
Array Option
3
RF Power Amplifier
2 kW BB-Amplifier
Magnet Supply and Monitor
SHIM
Gradient Power Amplifier
EPI Booster
Filter-plate
RF Application System
4
3
RF Coil Control
Air Comp. and Chiller Unit
Tuning
Shim Coil
Gradient Coil
(watercooled)
Body Coil
Surface Coils
Vision Options
RF Cabin

In the blockdiagram at left, the position of the options is shown.

- MRSC
  A satellite console can be connected via the network connection.
  The configuration is similar to the main console MRC. In principle the same software is running on this console, so that under the same user interface the same evaluation functions are available.

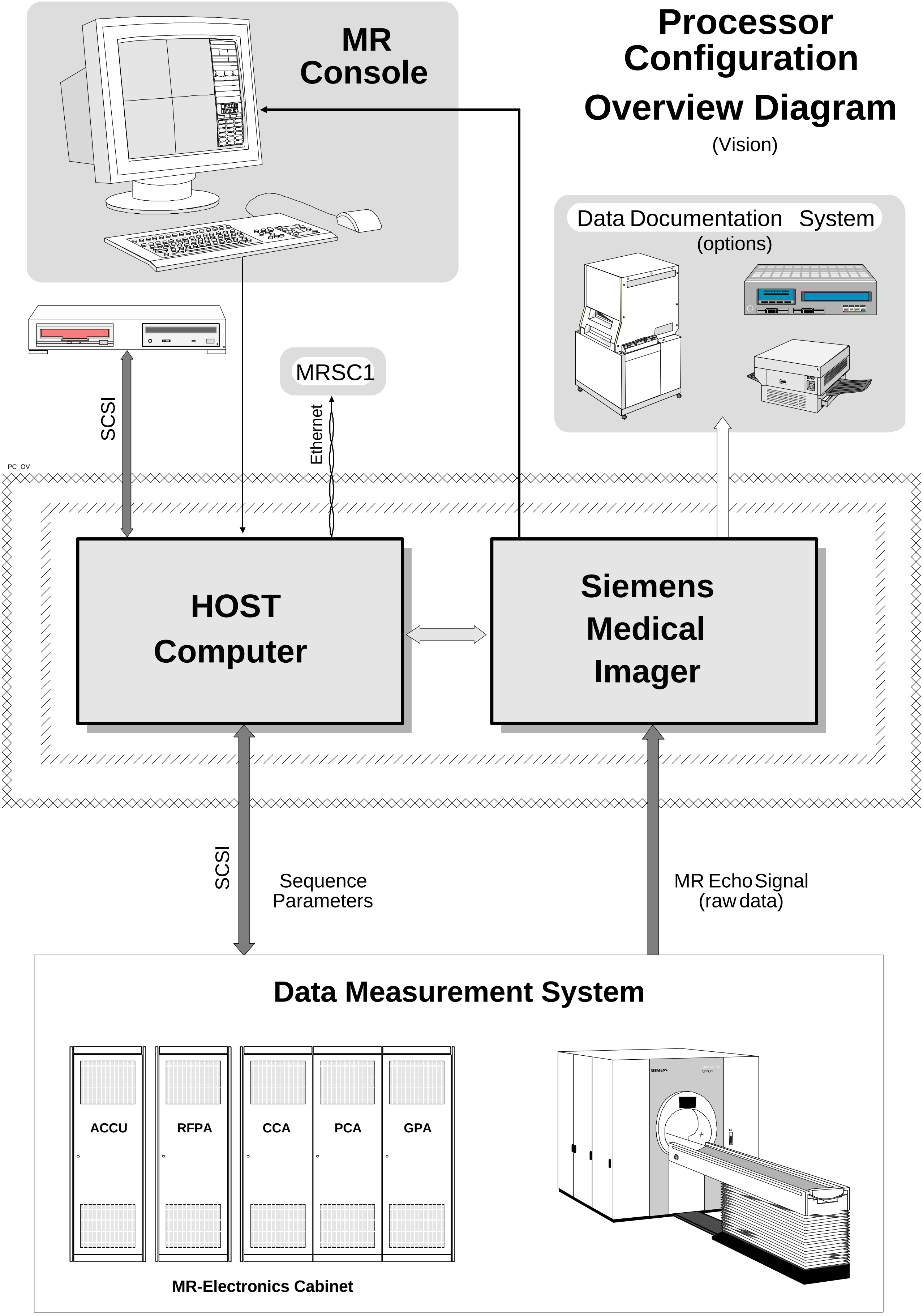
- Digital Camera
  For hardcopy documentation a camera can be connected via
  - a proprietary hardware interface (MAXICAM) or
  - a network connection using the DICOM Basic Print protocol

- Surface Coils
  A vatiety af surface coils is available to give optimum image quality for all applications.

- Array option
  For the available Array coils additional hardware (demodulator, LPF, ADC) is needed to allow simultanious data acquisition for four coils.

- EPI Booster
  The Gradient overdrive option allows faster seqences, since it increduces the Gradient rise time from 600us to 300us. It consists of a separate cabinet (Booster) and the control board ATG in the CCU.

- P31 Spectroscopy
  Since phosphorus has a resonance frequency of 25.66 MHz, extra hardware is required:
  - 2 kW broad band amlifier
  - RF coil for 25.66 MHz
  - a modified filter-plate
  - broad band T/R switch


MR
Console
Processor
Configuration
Overview Diagram
(Vision)
Data Documentation System
(options)
MRSC1
SCSI
Ethernet
PC_OV
HOST
Computer
Siemens
Medical
Imager
SCSI
Sequence
Parameters
MR Echo Signal
(raw data)
Data Measurement System
ACCU
RFPA
CCA
PCA
GPA
MR-Electronics Cabinet

Processor Configuration

The Processor Configuration (PC) consists of those components which provide the overall system control, image reconstruction and evaluation hardware and software, storage and archiving devices and various I/O interfaces. The main components comprising the PC are:

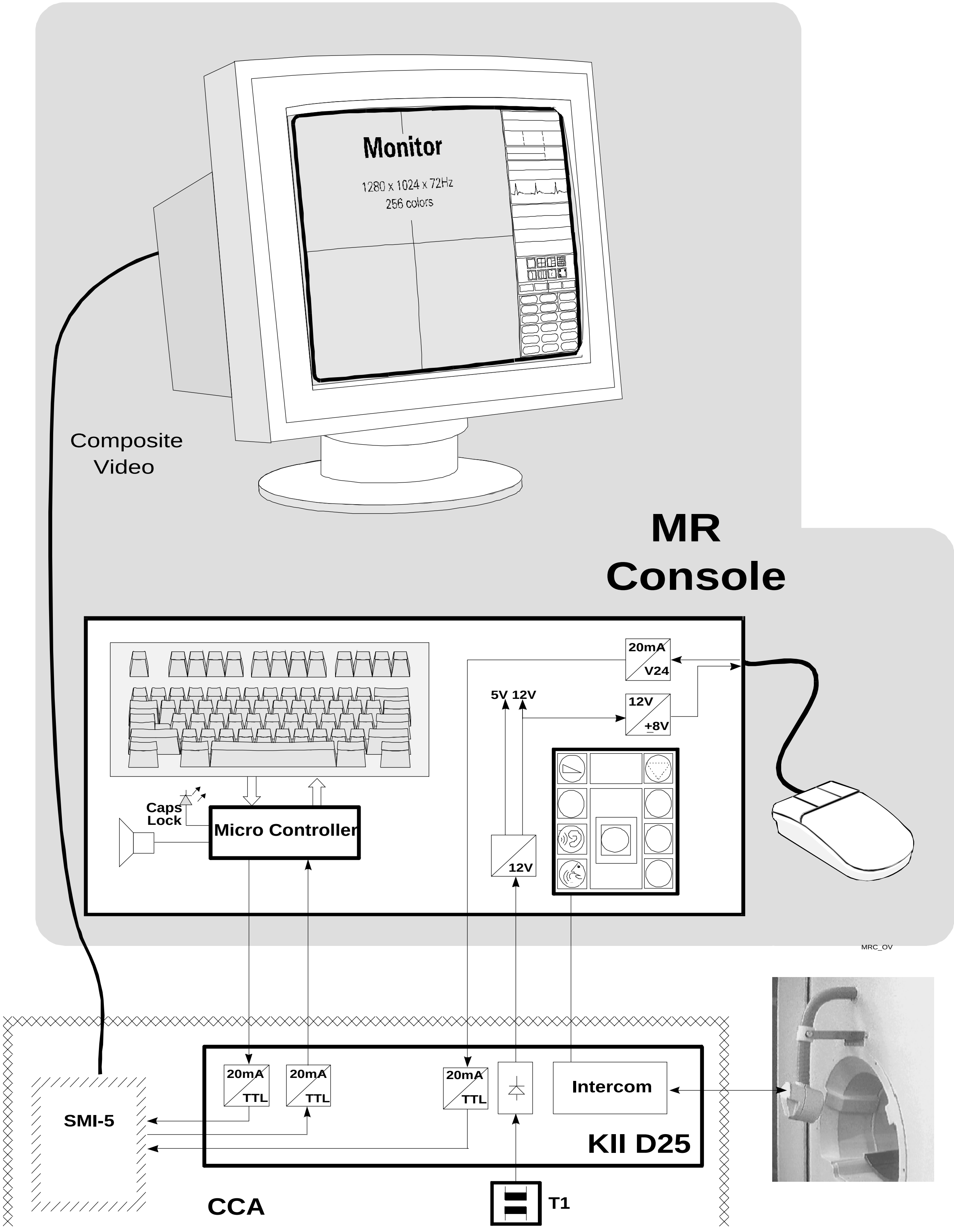
- MR Console
- HOST Computer
- SMI-5 Imager Electronics
- Data Documentation System

The MR Console provides user input functions via a keyboard and mouse. The software is fully menu driven which offers the convenience of 100% mouse control. Most program interfaces also provide for dialog input via keyboard entry for those who prefer to switch to the button pushing operation mode every now and then (good for the elbow bones).

Once the user has selected the sequence and parameters, the HOST generates the sequence specific hardware parameters and transfers them to the Data Measurement system (a SCSI device). After the sequence is performed, the acquired image data (raw data) is passed on to the SMI-5 which then calculates the image. The image can be further processed and archived to mass media (MOD or hard disk) and output as hardcopy images to a digital camera or printer.

Hardware Testing

The Host and SMI-5 components can be extensively tested with special diagnostic programs available through the service platform under NUMARIS. The Host and SMI-5 components contain built-in diagnostic routines which test hardware functionality. The diagnostics are varied in coverage and intensity which range from quick checks of vital functions to in-depth testing of memory and hardware ports. The Service platform serves as a centralized user interface to the individual component diagnostic routines. Through the service platform, the service engineer can select and run individual diagnostic test levels separately for each board component or blockwise to test a wide range of components simultaneously. This flexibility allows for discrete testing.


Monitor
1280 x 1024 x 72Hz
256 colors
Composite Video
MR Console
20mA
V24
5V 12V
12V
+8V
Caps Lock
Micro Controller
12V
MRC_OV
20mA
TTL
20mA
TTL
20mA
TTL
Intercom
SMI-5
KII D25
CCA
T1

MR Console

The MRC consists of the major user I/O devices which include :

- a high resolution **monitor** for image and dialog display
- a **keyboard** and **mouse** as input devices
- **MOD** and **CD** mass storage devices for archiving and software installation

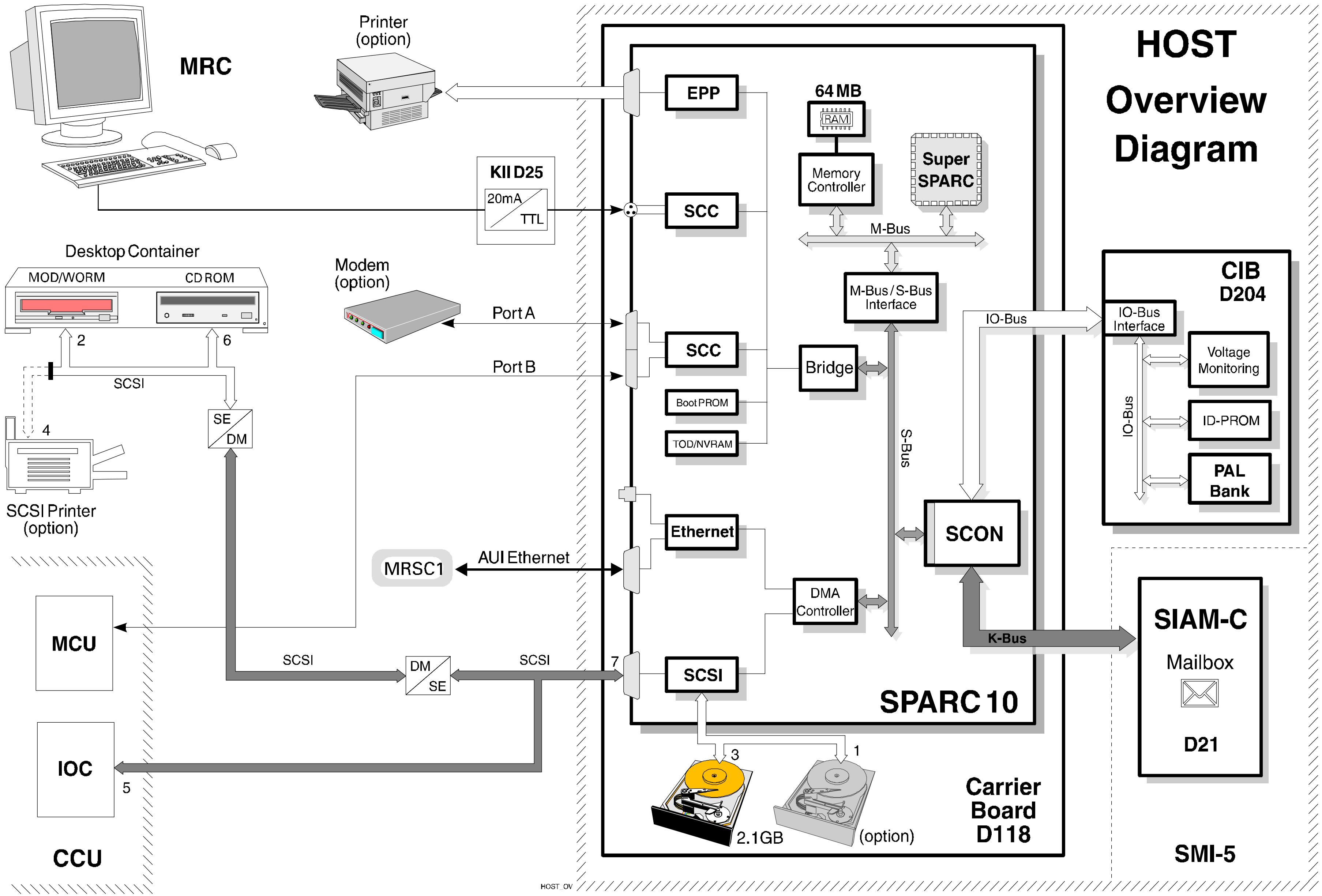
The keyboard and mouse input devices connect to the Host computer via the Keyboard & Intercom Interface board. The system's graphic card which generates the video signal for the monitor resides in the SMI-5. The mass storage devices are interfaced directly with the Host processor over an industry standard SCSI bus.

Intercom

An intercom system has been integrated in the keyboard for communication between the operator and the patient. Music may be piped into the examination room over the intercom system via an audio cassette or CD player when connected.

MR Satellite Console

An optional MR console (satellite console) can be installed as a parallel evaluation workstation. It contains the necessary hardware to process and perform evaluation functions on MR images. Two variations of the MRSC are available which allow either an installation to within a distance of 23 or 90 meters.


HOST
Overview
Diagram
MRC
Printer
(option)
KII D25
20mA
TTL
Desktop Container
MOD/WORM
CD ROM
2
6
SCSI
4
SE
DM
SCSI Printer
(option)
Modem
(option)
Port A
Port B
MRSC1
AUI Ethernet
MCU
IOC
5
CCU
SCSI
DM
SE
SCSI
7
EPP
SCC
SCC
Boot PROM
TOD/NVRAM
Ethernet
SCSI
64 MB
RAM
Memory
Controller
Super
SPARC
M-Bus
M-Bus / S-Bus
Interface
Bridge
S-Bus
DMA
Controller
IO-Bus
SCON
K-Bus
SPARC 10
3
1
2.1GB
(option)
Carrier
Board
D118
CIB
D204
IO-Bus
Interface
IO-Bus
Voltage
Monitoring
ID-PROM
PAL
Bank
SIAM-C
Mailbox
D21
SMI-5
HOST_OV

HOST

The HOST computer has at its core the SUN SPARCstation 10 and consists of the SPARCstation and the Cabinet Interface Board (CIB).

SPARC Processor

The SPARCstation is based on the Super **SPARC RISC** processor featuring superscalar pipelining and on-chip integer, floating point, memory management and cache controllers bringing about a very high processing capability. The CPU and memory components are attached over a dedicated processor bus (**M-Bus**) which is 64 bits wide and clocked at 40MHz. **64MB** of DRAM have been given to the processor as main memory. The on-board I/O peripherals are connected via an intelligent bus system (**S-Bus**) which is very comparable to the local bus systems in the PC world. The peripherals include :

- serial and parallel communication ports
- AUI (thick-wire) ethernet ports
  with SPARC 10 additionally twisted-pair
- SCSI bus interface
- ISDN and digital audio drivers

The S-Bus is also tached at 40MHz and is 32 bits wide. An SCON adapter interfaces the SPARC to the CIB and to the SMI-5 via the SIAM-C.

Carrier Board

The SPARC board is smaller then the SMI boards and is therefore mounted on a Carrier board which has the normal SMI board height. The system’s disk drive(s) is(are) also mounted on the Carrier board.

Cabinet Interface Board

The **CIB** acts as a universal interface which has the main job of housing the option PALs but also monitors supply voltages for the HOST and the SMI and provides a serial communication link adapter.

SCSI-Bus

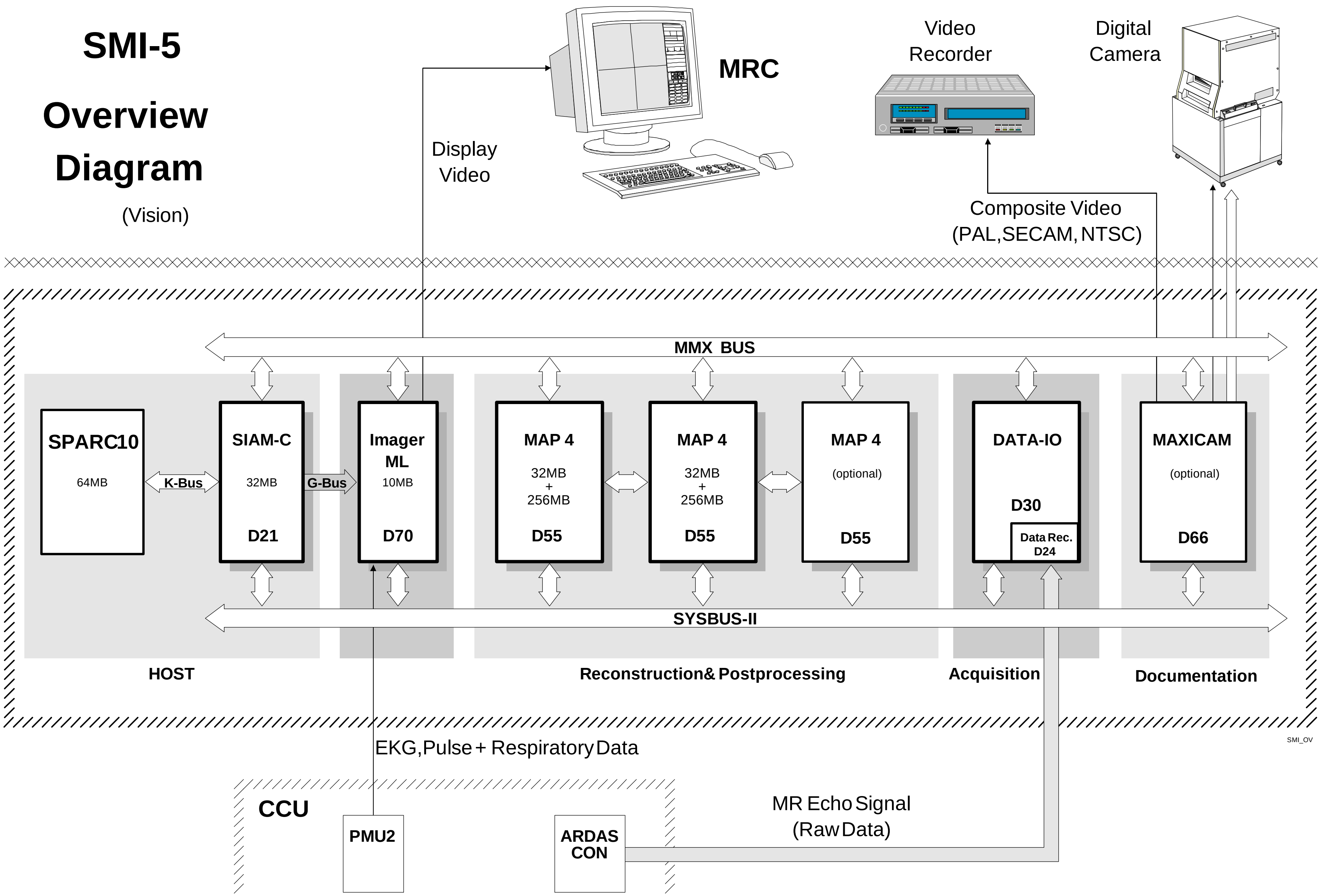
A modern **SCSI** (Small Computer System Interface) bus system connects the storage devices (hard disks, CD-ROM and MOD) and the Measurement System (via the CCU).

Storage Devices

A 2.1GB (giga byte) 3 1/2" **hard disk** carries the system software and image data. The image data storage capacity can be extended with the installation of a second hard disk (option). An **MOD** (magnetic-optical disk) serves as the archiving device. A **CD-ROM** drive is provided for loading and updating the software.

SMI-5 Overview Diagram

(Vision)



SMI-5

The SMI-5 is a specialized high-speed image calculation, processing and evaluation system providing image reconstruction, video display and hard-copy imaging functionality.

General Structure

Processing speed is not guaranteed with mere processor muscle. The processors must be supplied with data as fast as they can process it. To facilitate the transfering of large image files - 512x512x12=3MB! - a bus architecture capable of realizing a high data throughput and speed is required. The SMI-5 has been equipped with a bus system consisting of **two** major bus types :

- **SYSBUS** (SYStem BUS); for parameters and commands
- **MMX BUS** (Multiple MultipleX); for mass data transfers.

MMX BUS

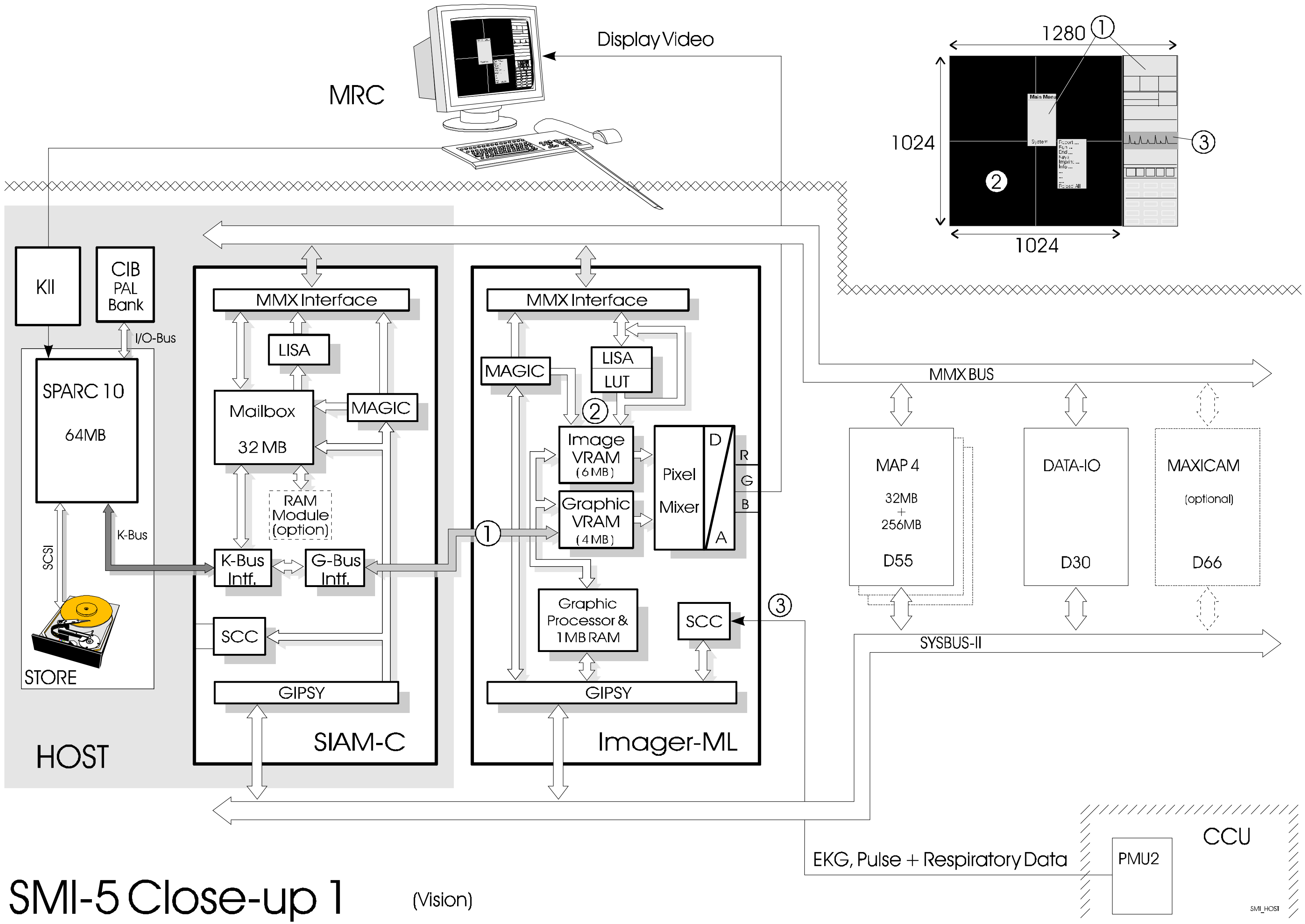
The **MMX BUS** consists of **four** 32-bit data busses. This bus is dedicated to transferring image data between the SMI components. Each bus is capable of transferring at a rate of up to 80MB/s. Theoretically, then, a peak data transfer of up to 320MB(!!) is possible. Each component connected to this bus possesses MMX interface logic containing a MAGIC bus controller. The **MAGIC** chip is a count-engine which acts as a fast DMA controller. It controls the addressing of the memory range to be read from or written to. The addressing parameters (bus select, address start, end and range values) for the transfer are supplied by the GIPSY.

SYSBUS

The **SYSBUS** is responsible solely for issuing commands and supplying parameters. As interface to this bus, each component is equipped with a **GIPSY** processor system. The GIPSYs are responsible for :

- SYSBUS interfacing (bus assignment and arbitration)
- communication
- providing the functionality of the component
- local hardware testing

During the boot of the SMI-5, the Host downloads the GIPSY software routines to the individual components via the SIAM-C, which acts as the overall bus master.



SMI-5 Close-up 1

(Vision)

SIAM-C

The SIAM-C (System Interface And Memory with Cable interface) is the interface between the Host and SMI-5. Communication between these units is handled via a "postal system". A large memory array (32MB) is set up as a "Mail Box" which allows anyone and everyone to communicate each other by sending and receiving mail. "Mail" consists of :

- run-time programs which must be down loaded to all SMI components
- look up tables, and calculation tables for image evaluation and display functions
- "datagrams" (commands, parameters)
- image files **from** STORE (hard disk) or images freshly calculated or modified by evaluation software to be stored **to** the hard disk

If the Host has sent mail to the SMI-5, the SIAM-C's GIPSY (also the SYSBUS Master) is responsible for determining who the mail is for and delivering it over the SYSBUS to the respective addressee. When dealing with images, the SIAM-C GIPSY simply informs the addressee that a BIG package has been received and arranges for a DMA exchange over the MMX BUS so that the addressee can pick up the mail himself.

In addition, the SIAM-C has been equipped with a Linear Interpolation and Shrink ASIC (**LISA**) which provides such functions as:

- sizing and windowing of images in preparation for camera exposure or printing
- general image display and simple evaluation functions such as shadow. zoom and windowing

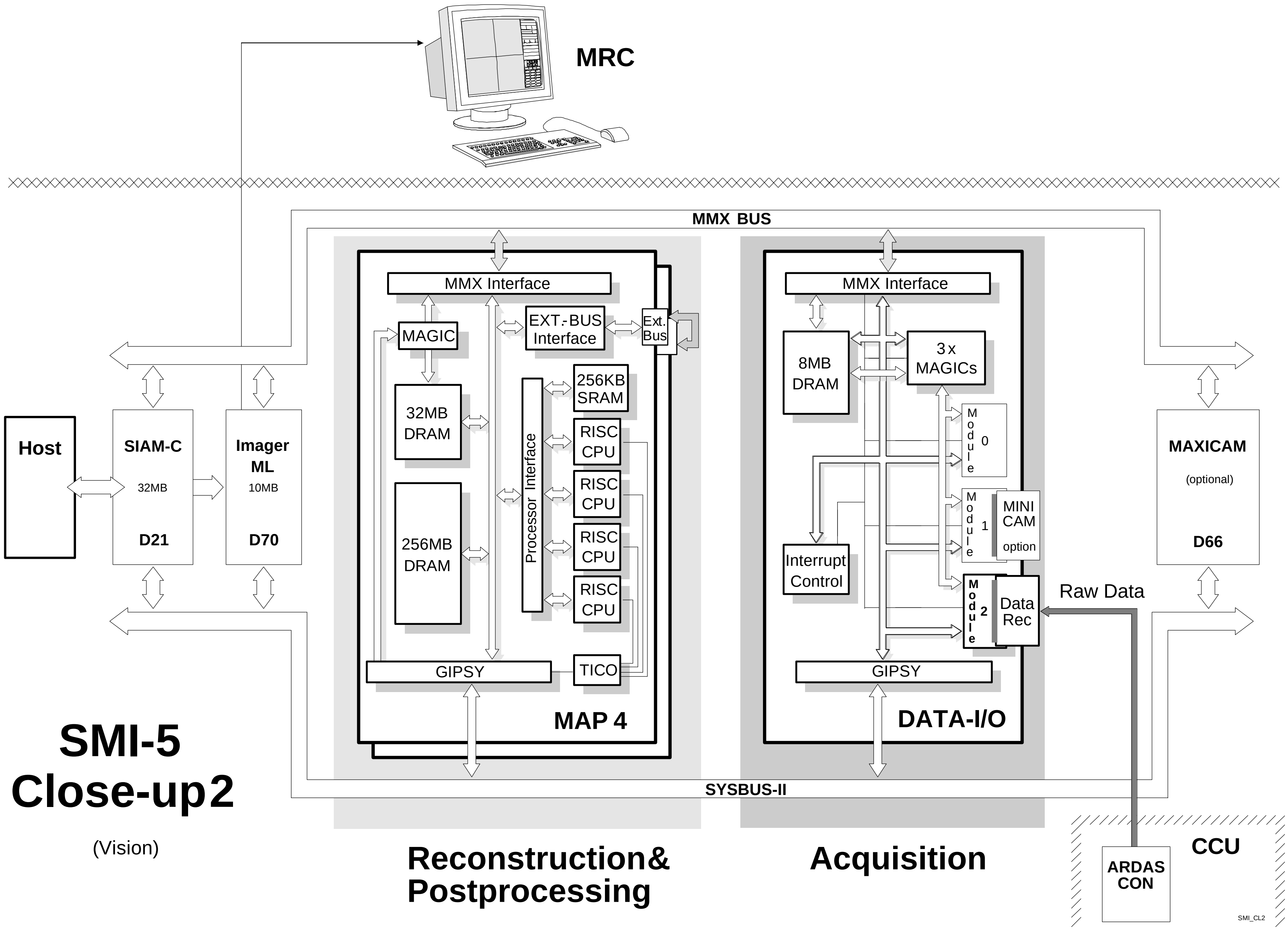
IMAGER ML

The IMAGER-ML is analogous to a graphic card in a personal computer but offers two big advantages. First, the structure of the IMAGER-ML's video RAM, where the data to be display on the screen is stored, is multi-ported. That allows the SMI-5, Host, or PMU to update portions of the monitor screen as required without waiting. The second major advantage is LOTS OF RAM (>10MB)!
The monitor display consists of images and graphic/text overlays from NUMARIS (menus, UNIX windows, etc.)

- The graphic display data generated by NUMARIS is delivered to the Imager-ML over the G-Bus.
- image data is piped in over the MMX BUS by the MAP or SIAM-C boards.
- Physiological data (ECG, respiratory, and pulse) from the PMU.

SMI-5 Close-up2

(Vision)



MAP 4

The processing muscle for pre- and post image processing and evaluation are found under the hood of the MAP4 boards. Each board (up to three boards maximum) possess 4 RISC processors in a special processor array set up for parallel processing. Vast amounts of RAM are available which further increase processing power and speed. The MAP4 (Memory And Processors) boards are responsible for :

- providing storage for pre-processed raw data
- performing the pre-processing functions (FFT, filtering)
- performing a wide range of post-processing functions: - multi-planar (3D) reconstruction (MPR)
  - maximum intensity projection (MIP) angio functions
  - cine, cardiac functions
  - zooming, roaming, scrolling, and magnification functions
  - profile, angle, distance measuring functions
  - flow quantification and mean value and statistical evaluations
  - image marking: grid/graphics drawing and text labeling
- buffering (image cache)

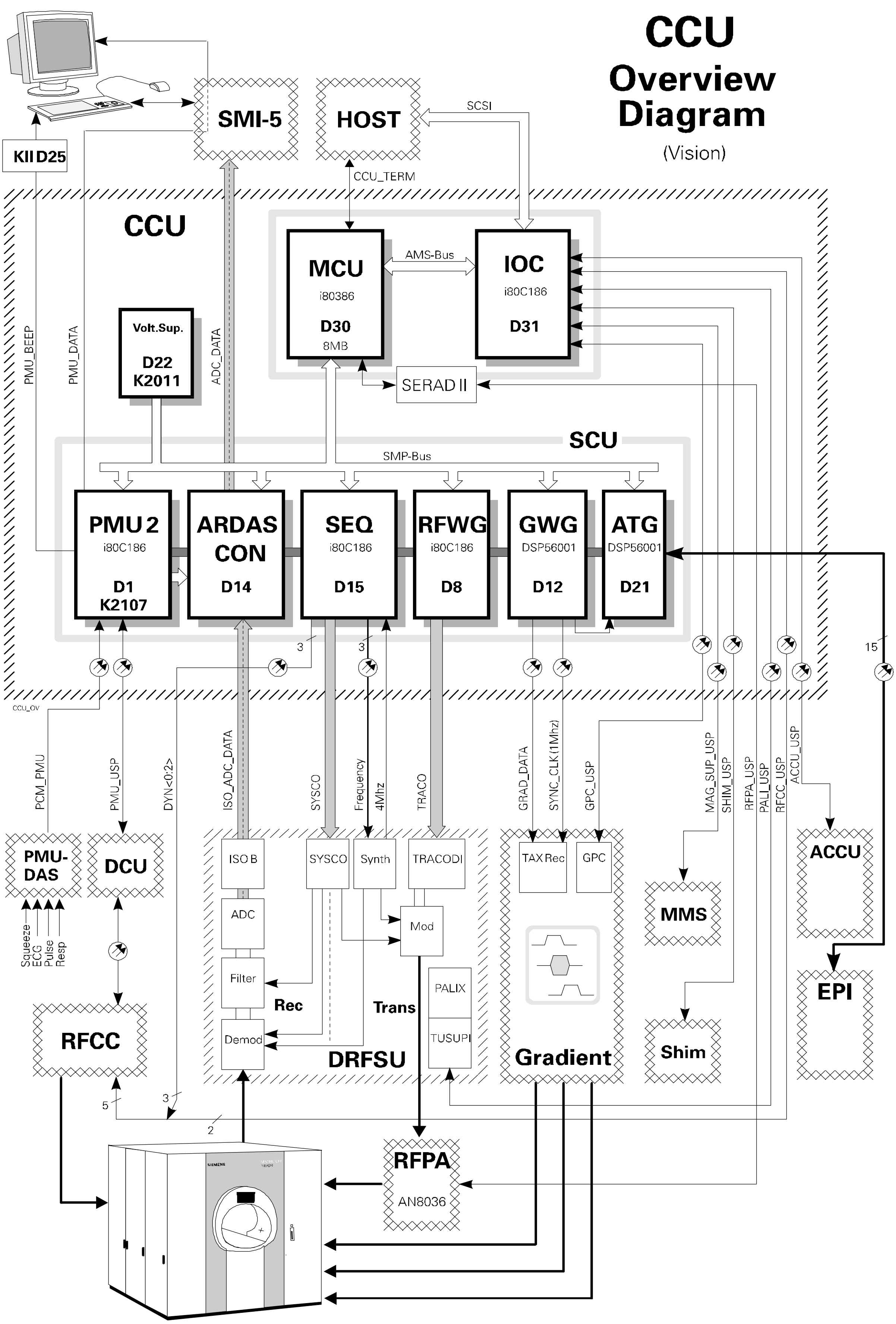
DATA-I/O

The DATA I/0 serves as the interface for piping the acquired MR data (raw data) from the data acquisition system (DAS) onto the internal SMI-5 busses. It provides three channels for plug-in modules each controlled by it’s own MAGIC processor for DMA control resulting in high data throughput. 8MB of RAM provide a sizable buffer accessible by all parties. Among it’s responsibilities are:

- control data transfer (receive data bursts from the DAS)
- buffer measurement data, i. e. receiving memory resides on the DATA I/0 board
- control data transfer between the DATA I/0 and MAP4 boards

MAXICAM

The MAXICAM provides the interface to photograph digital image data on High Resolution Laser Cameras and/or Standard Video Recorder (PAL, NTSC, SECAM) to record cine mode sequences in real time. Several different cameras are presently supported with the list growing continually. Siemens also offers a network configuration allowing photo archiving cameras to be used by different imaging system modalities produced by Siemens.



CCU
Overview
Diagram
(Vision)
SMI-5
HOST
SCSI
KII D25
CCU_TERM
CCU
MCU
i80386
D30
8MB
AMS-Bus
IOC
i80C186
D31
Volt.Sup.
D22
K2011
PMU_BEEP
PMU_DATA
ADC_DATA
SERAD II
SCU
SMP-Bus
PMU 2
i80C186
D1
K2107
ARDAS
CON
D14
SEQ
i80C186
D15
RFWG
i80C186
D8
GWG
DSP56001
D12
ATG
DSP56001
D21
3
3
15
CCU_OV
PCM_PMU
PMU_USP
DYN<0:2>
ISO_ADC_DATA
SYSCO
Frequency
4Mhz
TRACO
GRAD_DATA
SYNC_CLK(1Mhz)
GPC_USP
MAG_SUP_USP
SHIM_USP
RFPA_USP
PALI_USP
RFCC_USP
ACCU_USP
PMU-
DAS
DCU
Squeeze
ECG
Pulse
Resp
ISO B
ADC
Filter
Demod
SYSCO
Synth
TRACODI
Mod
PALIX
TUSUPI
Rec
Trans
DRFSU
TAX Rec
GPC
Gradient
MMS
Shim
ACCU
EPI
RFCC
5
3
2
RFPA
AN8036

Control and Communication Unit

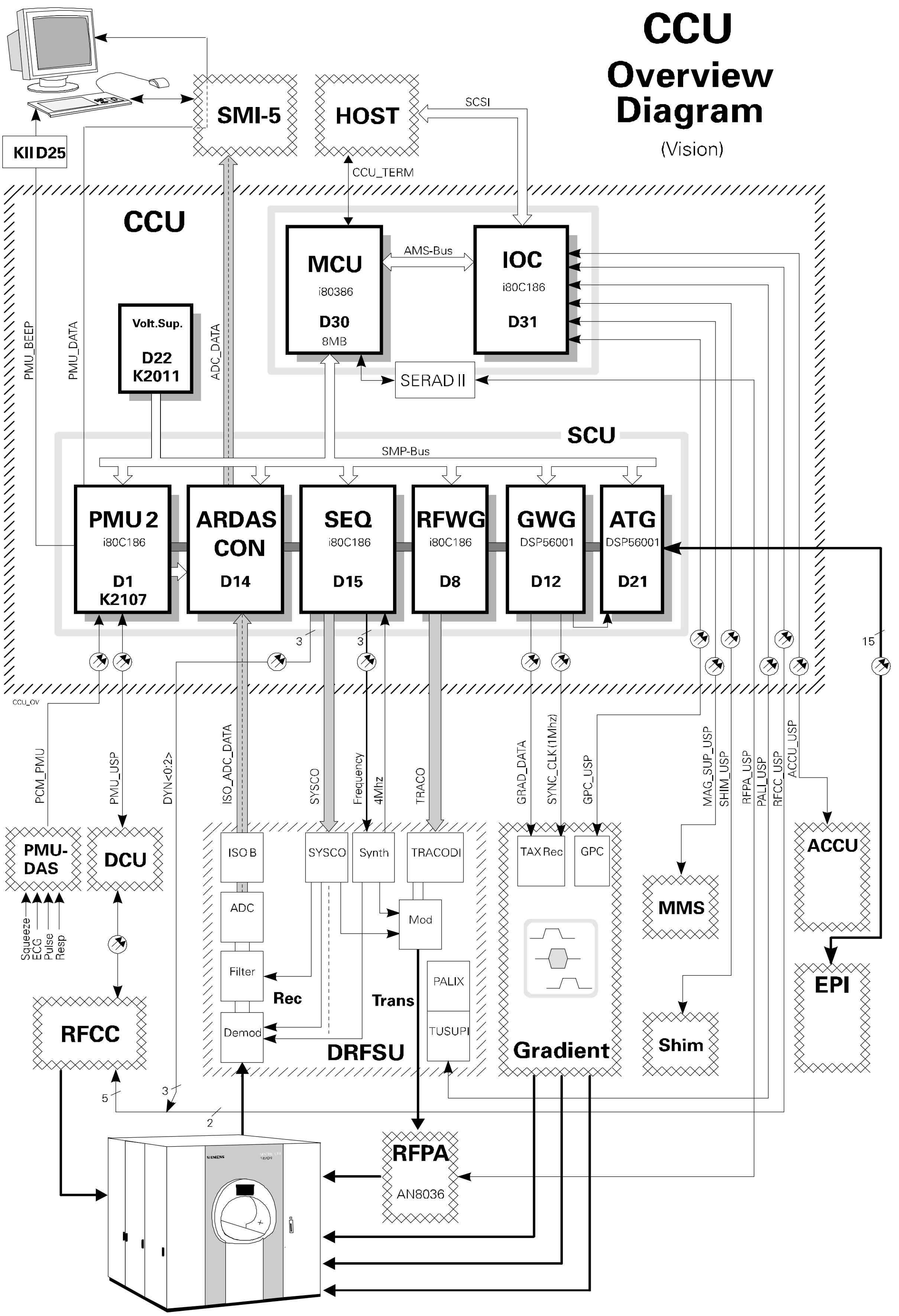
The CCU acts as the heart of the data measurement system. It is equipped with several microprocessor based control and signal processing boards which exercise control of all measurement system components. Its components can be placed into two main groups according to their tasks:

- Communication
- Sequence Control

As the main overseer, the MCU (main control unit) has the responsibility of delegating the communication and sequence control operations. It controls the down-loading of the software to the SCU components and generally keeps house by satisfying any data needs and keeping tabs on the overall status.

Hardware Testing

Special test software has been written which can be down-loaded into each of the SCU components. These tests cover internal hardware checks and external connections with the help of loop-back connections which are manually connected. The tests can be configured to analyze various hardware layers (memory tests, I/O tests, discrete functional blocks particular to each board, etc.) on an individual basis providing freedom and flexibility in trouble shooting.


CCU
Overview
Diagram
(Vision)
SMI-5
HOST
SCSI
CCU_TERM
KII D25
CCU
MCU
i80386
D30
8MB
AMS-Bus
IOC
i80C186
D31
SERAD II
Volt.Sup.
D22
K2011
PMU_BEEP
PMU_DATA
ADC_DATA
SCU
SMP-Bus
PMU 2
i80C186
D1
K2107
ARDAS
CON
D14
SEQ
i80C186
D15
RFWG
i80C186
D8
GWG
DSP56001
D12
ATG
DSP56001
D21
CCU_OV
PCM_PMU
PMU_USP
DYN<0:2>
ISO_ADC_DATA
SYSCO
Frequency
4Mhz
TRACO
GRAD_DATA
SYNC_CLK(1Mhz)
GPC_USP
MAG_SUP_USP
SHIM_USP
RFPA_USP
PALI_USP
RFCC_USP
ACCU_USP
PMU-
DAS
DCU
Squeeze
ECG
Pulse
Resp
ISO B
ADC
Filter
Demod
Rec
SYSCO
Synth
TRACODI
Mod
PALIX
TUSUPI
Trans
DRFSU
TAX Rec
GPC
Gradient
MMS
Shim
ACCU
EPI
RFCC
RFPA
AN8036

Communication

The IOC (I/O controller) provides the MCU with external communication links to both the Host and MR measurement system components.

The IOC provides the main communication path to the Host which supplies the SCU with:

- run-time software (loadware)
- sequence parameters and data
- carry out tests of SCU hardware and communication lines

The Data Measurement sub-systems are each equipped with a microprocessor system for controlling component specific functions and internal monitoring. A list of the major software/hardware parameters and status information being passed can be summarized:

- **MMS** (Magnet Monitoring System)

  Status : LHe fill level status, magnet shield temperature, ERDU battery charge

- **SHIM**

  Commands: downloading of shim currents
  Status: Overtemperature, Overcurrent

- **ACCU** (Air Compressor and Chiller Unit)

  Commands: remote power on/off commands for water chiller
  Status: cold-head compressor halted (error)

- **RFCU** (RF Coil Control Unit - in RFCC)

  Commands: coil tuning data, coil control parameters, remote loop test commands for coil control electronics and patient table
  Status: coil code, tune request, test status and error information

- **TUSUPI** (Tuning and Supervision Interface - in DRFSU)

  Commands: PALI-SAR parameters for patient monitoring
  Status: RF hardware faults such as excessive reflected power,
  excessive forward or peak power (with multi-level power monitoring provided depending on coil type)

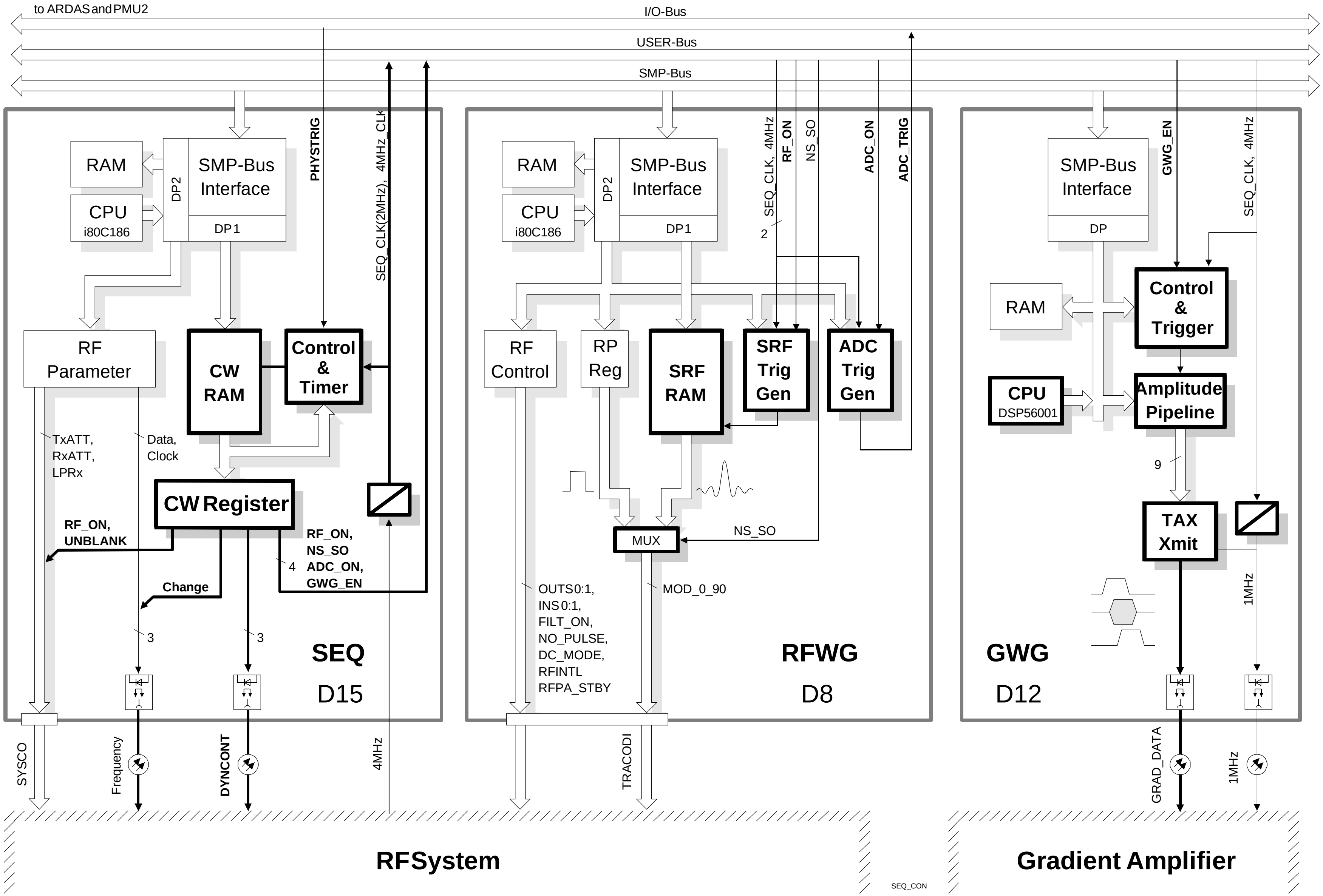
- **GPC** (Gradient Processor Control)

  Commands: gradient pulse delay times
  Status: regulation errors, overcurrent, duty cycle, coil and amplifier over temperature information.

- **RFPA** (RF Power Amplifier)

  Commands: power_on/off, reset, status request, tube hour request.
  Status: Peak power, high reflected power, temperature, overcurrent errors

Sequence Control Unit



Sequence Control Unit

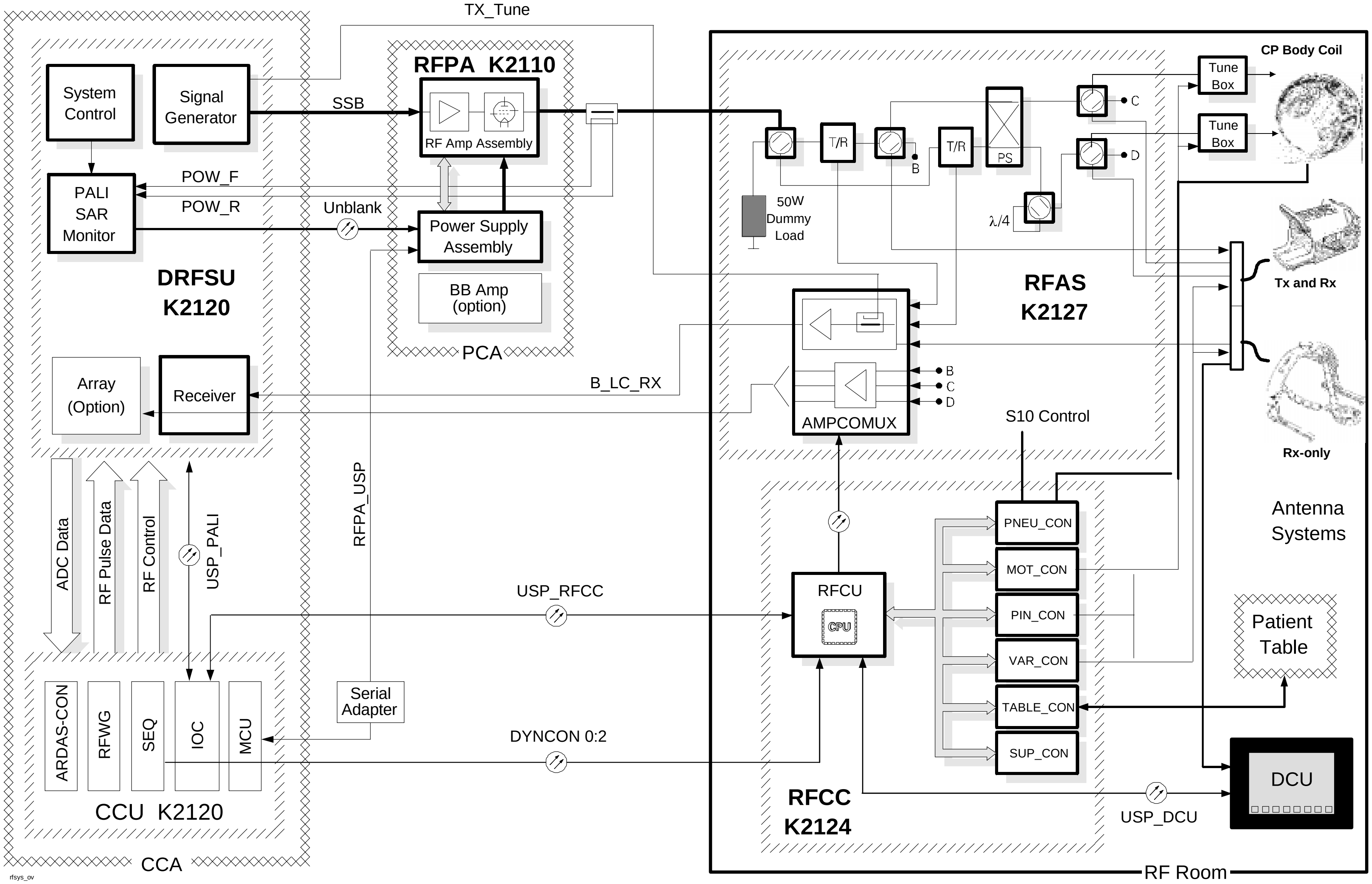
The components which comprise the Data Measurement System are the:

- RF System
- Gradient System
- Shim
- Data Acquisition System

It is the task of the SCU hardware to orchestrate the Data Measurement System components during the imaging sequence. The SCU provides the RF and gradient signal waveforms and generates the timing signals for synchronizing the RF, gradient and data acquisition (ADC) trigger pulses. Additionally, It performs the RF coil tuning, frequency and transmitter adjustments. The SCU components and their tasks can be summarized as follows:

- **SEQ**
  Coordinates the overall timing of the above mentioned components and the RF system components. It also selects transmit and receiver attenuator values, lowpass filter cutoff frequency setting as well as the carrier frequency of the synthesizer.
- **RFWG**
  Supplies the RF pulse envelopes and static control signals for the RF signal generator and receiver. Generates the ADC trigger for data acquisition.
- **GWG**
  Supplies the gradient amplifier with the gradient pulse waveforms for all three axis. It's digital signal processor is used to calculate the coil tuning results and the frequency and transmit adjustment results.
- **ARDAS_CON**
  Provides an interface between the data acquisition system (ADC) and the SMI-5. It can modify the ADC trigger to provide ECG or respiratory gating (retrogating) functions. When the ARRAY option is installed, the ARDAS_CON provides MUX functions for the ADC trigger and acquired ADC data for the four receive channels.

Once the sequence parameters have been selected, the HOST down-loads the sequence data to the MCU which distributes the RF and pulse amplitudes to the RFWG's SRF RAM, the gradient pulse table to the GWG and the sequence timing to the SEQ's CW-RAM. With a start of the sequence, the RF and gradient signals will be generated and sent to the DRFSU and GPA respectively.



RF System Overview Diagram (Vision)

RF System

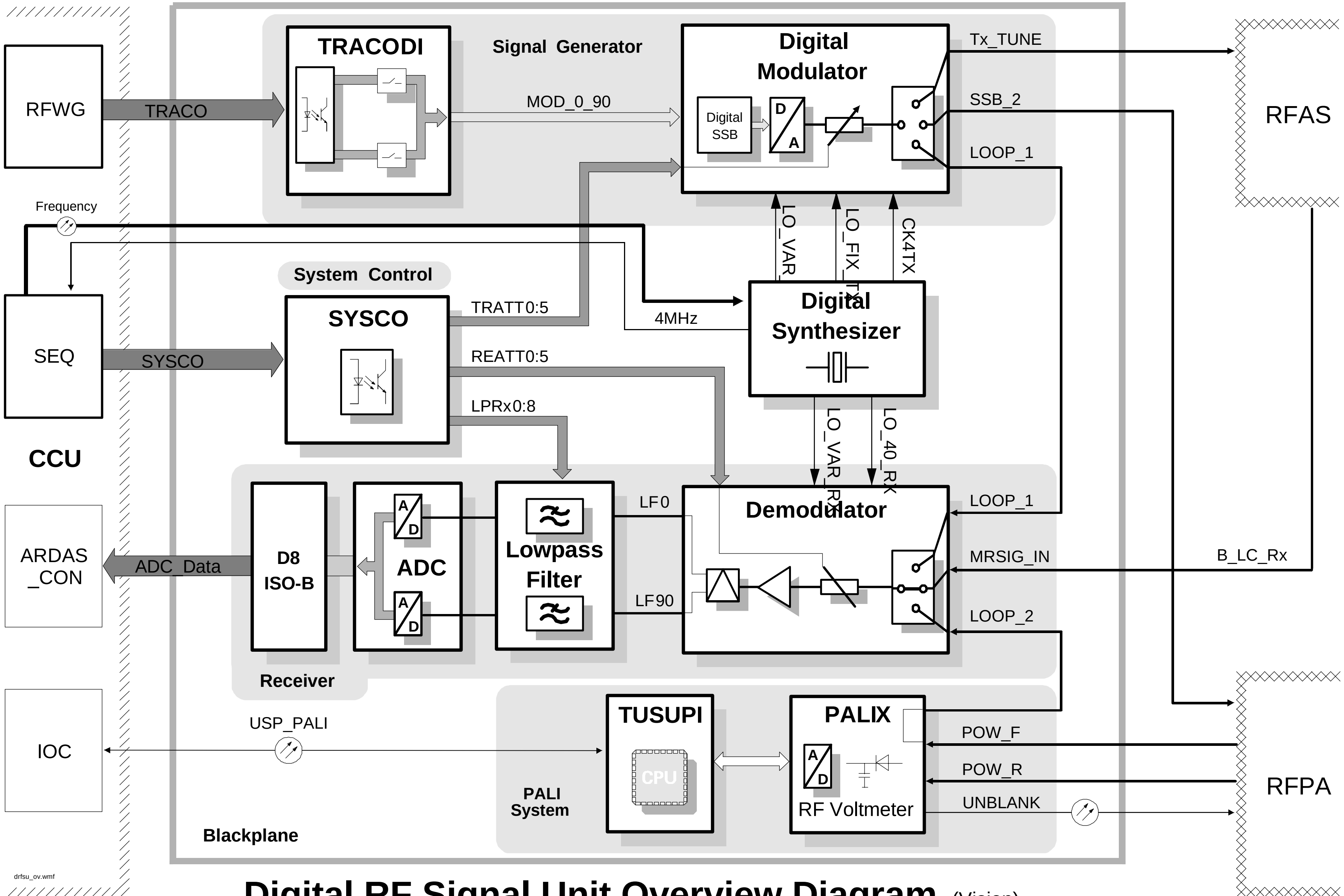
The RF System is responsible for generation and transmission of the excitation pulses which define the slice profile (position and thickness of the slice) according to the region to be imaged. After excitation, various gradients are applied which will result in an MR echo signal. The echo must be picked up (received) by a receive antenna, amplified and processed (i.e. demodulated, filtered and digitized) for preparation of the image calculation and display.

The hardware of the RF system can be subdivided into 5 groups :

- Digital Radio Frequency Signal Unit (**DRFSU**)
- Radio Frequency Power Amplifier (**RFPA**)
- Radio Frequency Application System (**RFAS**)
- Radio Frequency Coil Control (**RFCC**)
- Antenna Systems

Hardware Testing

Built-in test loops, covering most of the RF System hardware components, have been incorporated into the RF System to facilitate testing and trouble shooting. Test sequences specially designed to test functionality and performance - linearity and stability - can be configured and started from the service software platform. Evaluation programs can then be evoked at the push of a button. Results are display graphically and alphanumerically with tolerances and performance data being given.
Since most tests require no further connections to the original configuration, it will be possible with future software interfaces that these tests can also be evoked remotely via modem with little or no human intervention.



Digital RF Signal Unit Overview Diagram (Vision)

Digital RF Signal Unit

The Digital RF Signal Unit is a broad band computer-controlled digital signal processing unit which contains the hardware components for generating the RF excitation pulse and processing the MR echo signal. The DRFSU is located at the back of the CCA cabinet.

Signal Generator

The RF excitation signal generation takes place inside the **Modulator** cassette which produces the SSB (single side band) excitation pulse. The shape and amplitude information of the pulse are downloaded by the RFWG as digital values. The generation of the RF pulse utilizes a mixture of digital and analog processing technologies which take advantage of the extremely phase stability (crucial for MR) on the digital side and clean signal spectrum on the analog side. The RF output can be in a range between 15-85 MHz which makes spectral imaging with other nuclei possible.

Receiver

The receiver consists of the **Demodulator** module, Low-Pass Filter and the ADC (analog to digital converter). The demodulator removes the high frequency content from the MR-echo signal to extract the actual image information. The Demodulator cassette also employs a combination of digital and analog processing methods resulting in stable and high resolution image data.

Lowpass Filter

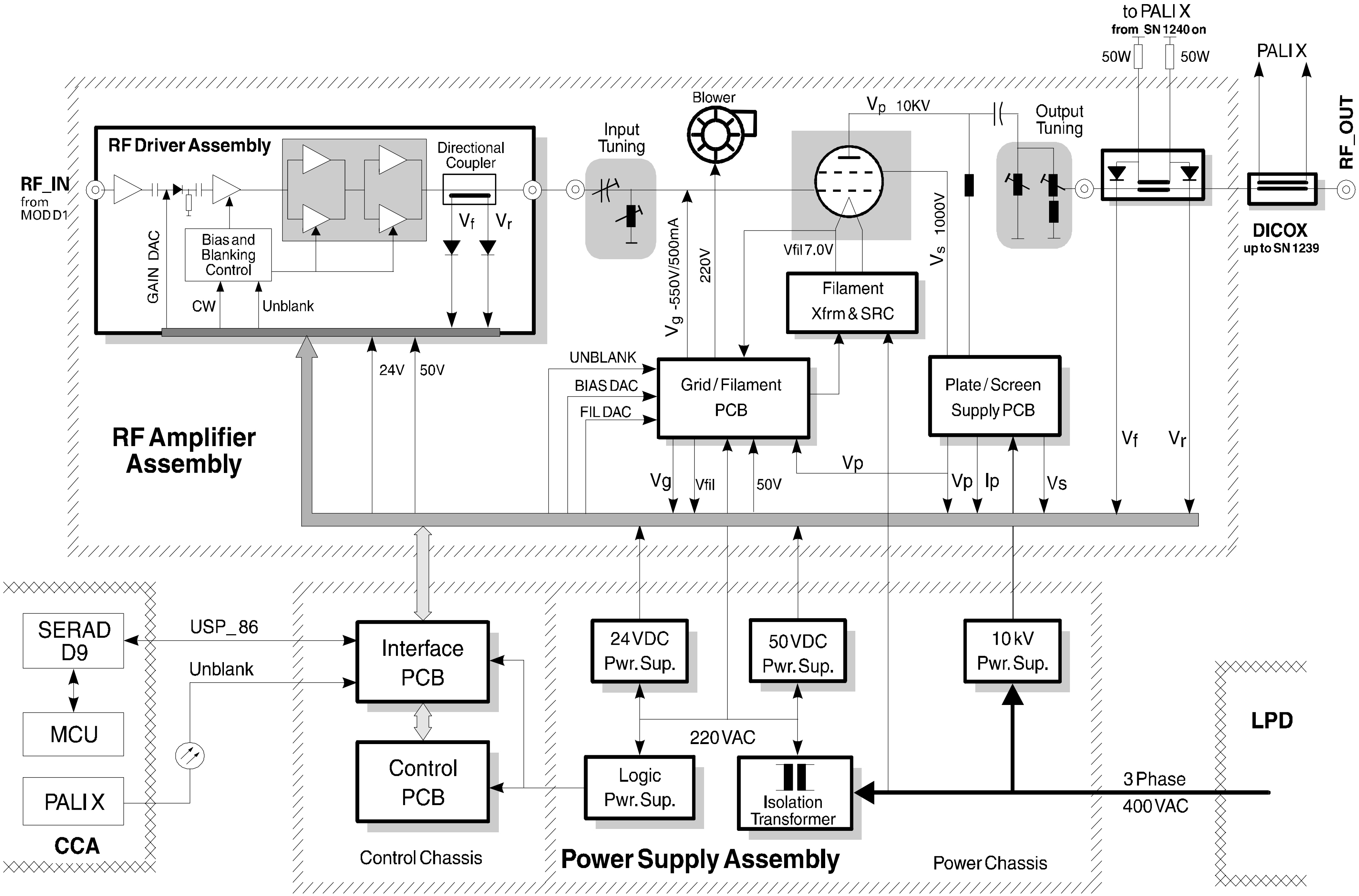
The **Lowpass Filter** reduces the distortion caused during the demodulation process and cuts out the frequencies outside of the FOV window to prevent fold-over artifacts.

Synthesizer

The **Synthesizer** is a broad band digital controlled synthesizer (15-85 MHz) which provides the modulator and demodulator with the required carrier frequencies for the modulation and demodulation processes.

PALI System

The **PALI System** provides a patient safety monitoring designed to prevent overheating due to excessive RF energy. The PALI also provides secondary hardware safety monitoring aimed at keeping the smoke to a minimum.



ANALOGIC AN8036S2 RF Power Amplifier

RF Power Amplifier

The Analogic AN8036S2 RF Power Amplifier is responsible for amplifying the RF pulse from the modulator to the power levels needed for excitation. M_0 vector flip angles of 180° for an average patient weight of 80 kilograms requires anywhere between 280 to 350 volts (depending on weight distribution) when using the body coil.

The RFPA must be stable during transmission and noise free during reception.

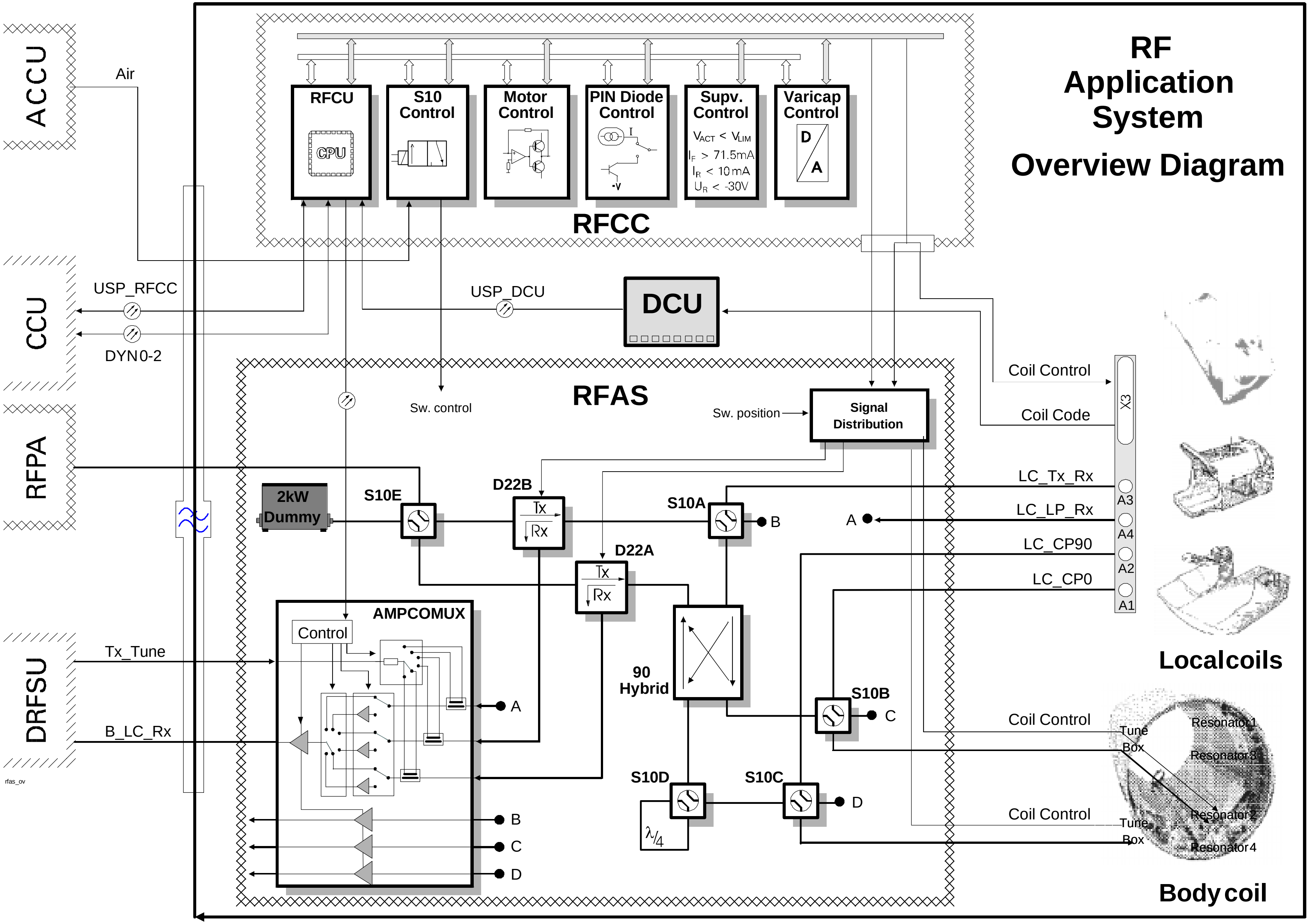
Functional Overview

The AN8036S2 is a two stage **class AB** amplifier with a solid-state driver amplifier and a tube based final stage power amplifier. This amplifier is used exclusively for proton imaging and has an operational frequency fixed at **63.6 MHz**. Maximum output power is rated at **15 kW** peak envelop power (PEP) at a maximum duty cycle (on versus off time) of 10%.

The amplifier employs a **microprocessor system** which exercises control over the amplifier and also monitors operational statistics. The extensive monitoring covers all operating voltages and checks for excessive plate current, pulse duty cycle, driver and tube output power, excessive standing waves, and tube temperature. Irreversible damage is therefore prevented should adverse conditions arise, such as over-driving the amplifier with excessive input power or pulse duty cycle.

The service platform loop tests can be used for making functional and performance tests on the RFPA. A laptop can be connected to the amplifiers control module giving access to the amplifiers monitoring circuits error and status messages. The control module software also contains commands offering test possiblities.

The amplifier can be field calibrated. The calibration procedure is supported by firmware. Calibration requires field-available test equipment (i.e., URV5 RF Millivolt meter, O'scope and DVM)



RF Application System
Overview Diagram
ACCU
Air
CCU
USP_RFCC
DYN0-2
RFPA
DRFSU
Tx_Tune
B_LC_Rx
RFCC
RFCU
CPU
S10 Control
Motor Control
PIN Diode Control
Supv. Control
$V_{ACT} < V_{LIM}$
$I_F > 71.5mA$
$I_R < 10 mA$
$U_R < -30V$
Varicap Control
D
A
DCU
USP_DCU
Sw. control
RFAS
Sw. position
Signal Distribution
2kW Dummy
S10E
D22B
Tx
Rx
S10A
D22A
AMPCOMUX
Control
90 Hybrid
S10B
S10C
S10D
$\lambda/4$
A
B
C
D
Coil Control
Coil Code
X3
LC_Tx_Rx
LC_LP_Rx
LC_CP90
LC_CP0
A3
A4
A2
A1
Localcoils
Tune Box
Resonator 1
Resonator 3
Resonator 2
Resonator 4
Body coil
rfas_ov

RF Application System

The RFAS is designed as a part of the front-end electronics for the 1.5T Vision System. It is responsible for applying the RF pulses and receiving the MR echo. In such a system where a wide variety of antenna types are employed, the RFAS must provide several functional components to support the various coil requirements.

Amplification

The received MR echo is extremely small and must be amplified. The method and type of amplification will play a very decisive role in the overall signal-to-noise quality of the receive system. The trick is to amplify the MR echo out of the noise floor without amplifying the noise as well. Several steps have been taken to optimize the **S/N**: distances between antenna and preamplifier have been kept short as possible; array coil types have built-in amplifiers; and amplifiers have been designed for an optimized signal-to-noise ratio using the best FET semi-conductor components the industry currently has to offer.

Array Mode

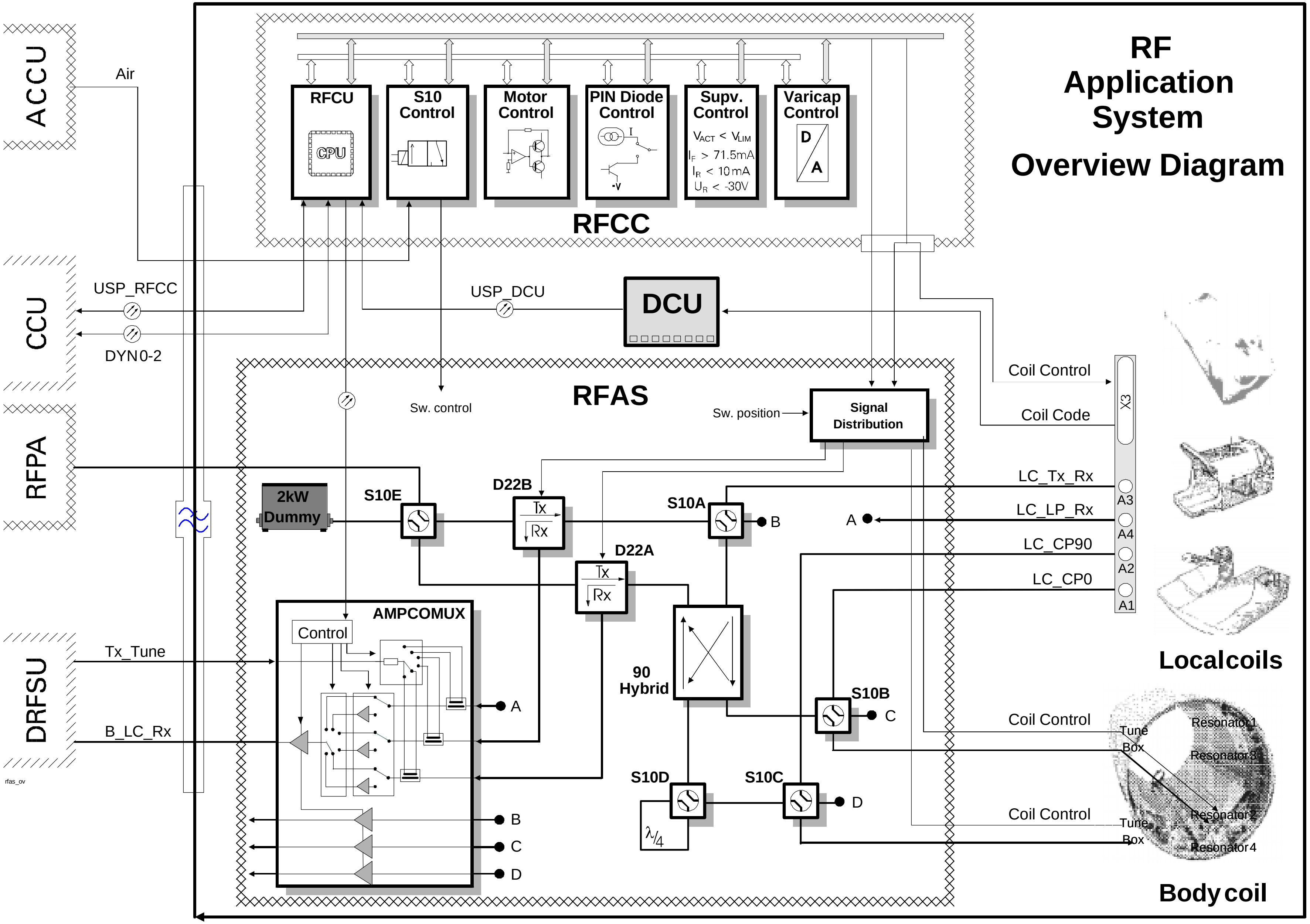
The RFAS has been prepared to handle the latest antenna designs: the Array coils. The **AMPCOMUX** houses the external amplifiers and provides for up to four receive channels. It has been designed to support all the antenna types planned for use on the Vision system. The amplifier channels can be bypassed to accommodate for antenna systems with or without built-in pre-amplifiers. The overall gain can also be controlled by amplifier stages which have variable gains (switchable).

Tx/Rx Switching

A **T/R Switch** is required when transmitting and receiving with a common antenna system. The switch must alternately couple the antenna to the transmitter during transmission and to the receiver during reception. The T/R switches implemented in the Vision are actively switched which assures a linear transfer characteristic (i.e., no feed-through distortion) and have high power handling capabilities.

D22B is used for transmit-receive LP local coils (e.g., spectroscopy coils). D22A is used to provide the 90° hybrid with a 50Ω termination during transmission (required if the hybrid is to work properly) and at the same time to dump the leakage current coming out of the hybrid.

The Hybrid acts as the transmit receive switch for the CP antenna systems.



RF Application System
Overview Diagram
ACCU
Air
CCU
USP_RFCC
DYN0-2
RFPA
DRFSU
Tx_Tune
B_LC_Rx
rfas_ov
RFCU
CPU
S10 Control
Motor Control
PIN Diode Control
Supv. Control
VACT < VLIM
IF > 71.5mA
IR < 10mA
UR < -30V
Varicap Control
D
A
RFCC
USP_DCU
DCU
RFAS
Sw. control
Sw. position
Signal Distribution
2kW Dummy
S10E
D22B
Tx
Rx
S10A
B
A
D22A
AMPCOMUX
Control
90 Hybrid
S10B
C
S10D
S10C
D
λ/4
Coil Control
Coil Code
X3
LC_Tx_Rx
LC_LP_Rx
LC_CP90
LC_CP0
A3
A4
A2
A1
Localcoils
Tune Box
Resonator1
Resonator3
Resonator2
Resonator4
Body coil

Tx/Rx Path Selection

The RFAS configuration allows for several transmit and receive paths according to the antenna or antenna systems being used. The following configurations have been provided for:

- Tx and Rx linear polarized Local coils (spectroscopy coils)
- Rx-only linear polarized Local coils (Oval Spine, Small FOV, Neck HH)
- Rx-only circular polarized Local coils (Spine Array, Body HH)
- Tx and Rx circular polarized Local coils (Head, Head/Neck, Body)

Air-controlled **coax switches** with extremely low insertion losses keep signal loss to a minimum while providing trouble-free operation in high-field magnetic environments.

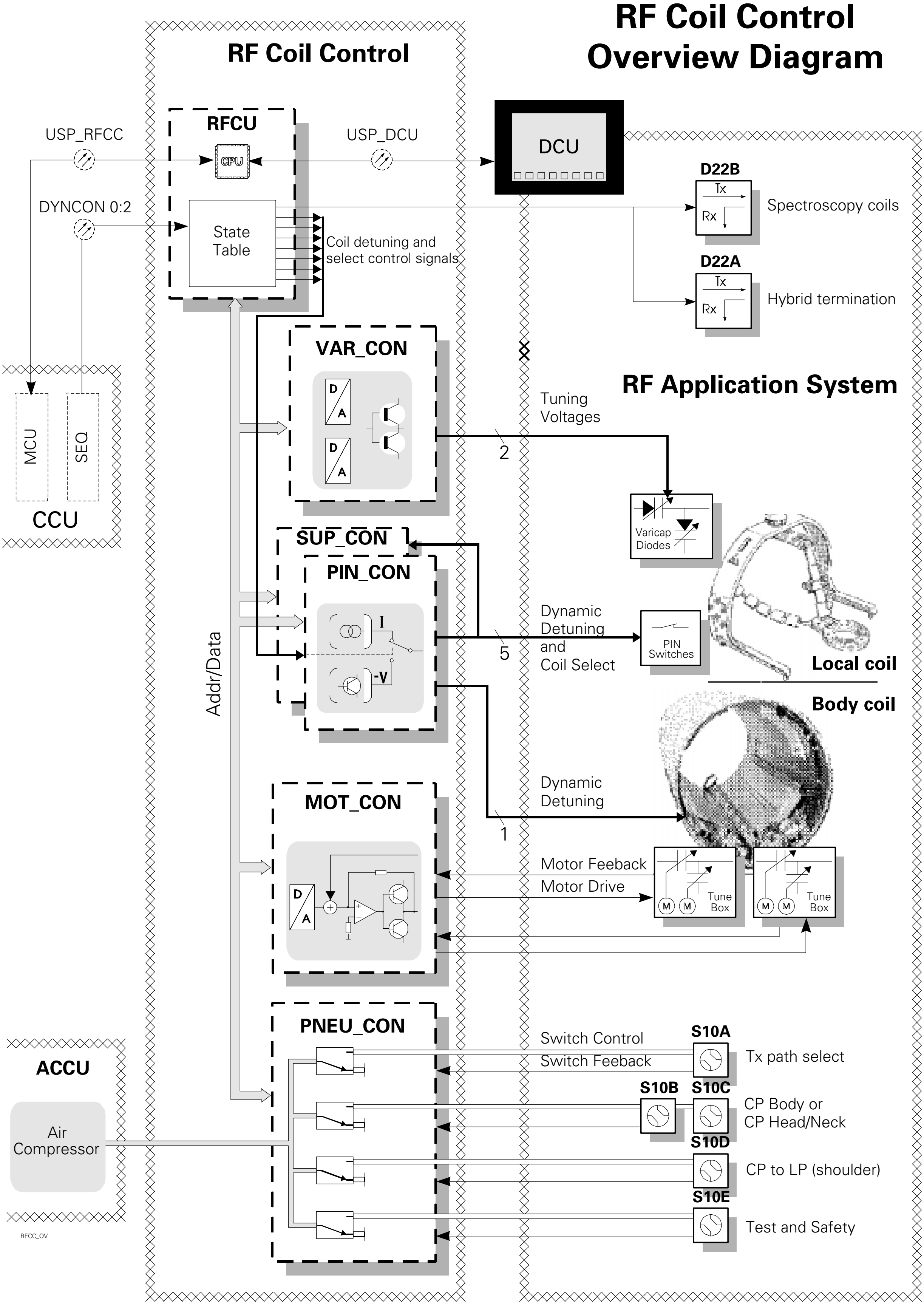
Tuning

Antenna tuning involves matching the antenna impedance to the transmission line impedance via an adjustable **matching network** (consisting typically of vacuum capacitors or varicap diodes) in order to maximize the transfer of power. Impedance matching (tuning) generally must be carried out whenever placing or repositioning the patient in respect to the antenna system, since the loading and load distribution can affect the antenna’s electrical impedance characteristics.

Depending on the design criteria and antenna loading attributes, tuning such antenna systems can be a complicated and lengthy process. A new tuning method, called Exact Tuning algorithm (**ET/a**) has been developed which performs the antenna tuning within seconds. Before the ET/a can be realized, a calibration of the matching network must first be carried out. The calibration "maps out" the complete tuning characteristics of the tuning elements and antenna hardware under varying load conditions. Once this calibration has been performed, it will be possible, after making an initial tune measurement, to use the tune data to calculate the necessary adjustments to the motors or varicap diodes to properly match the impedance.

Several coil systems (e.g. all Array antenna systems, all flex coil systems, CP Head coil, CP Extremity) have fixed matching networks and require no tuning.

RF Coil Control Overview Diagram



RF Coil Control

The RF Coil Control electronics provide the control signals for the RFAS and antenna components and implements several hardware monitoring features designed to protect man and machine.

Dynamic De-tuning

When using separate transmit and receive coils it becomes necessary to switch between coils "on-the-fly". PIN diodes are used as electronic switches. They are strategically placed inside the antenna assemblies in a way that the antennas can be tuned (enabled) or de-tuned (disabled) simply be opening or closing these diode switches. The PIN diodes are controlled by the **PIN_CON** in the RFCC.

The PIN diodes play a vital role in coil control and therefore are pivotal in assuring patient safety. RF burns can be caused if localized antennas are not properly controlled. The **SUP_CON** gives an added measure of safety by monitoring the PIN currents and voltages.

Coax Switch Control

The coax switches of the RFAS operate with air pressure. Pressurized air (6-8 bar) is supplied to the individual coax switches through microprocessor controlled solenoid valves. Switch states are verified with feedback signals from built-in position detection micro switches. Before RF can be applied, switch states are verified.

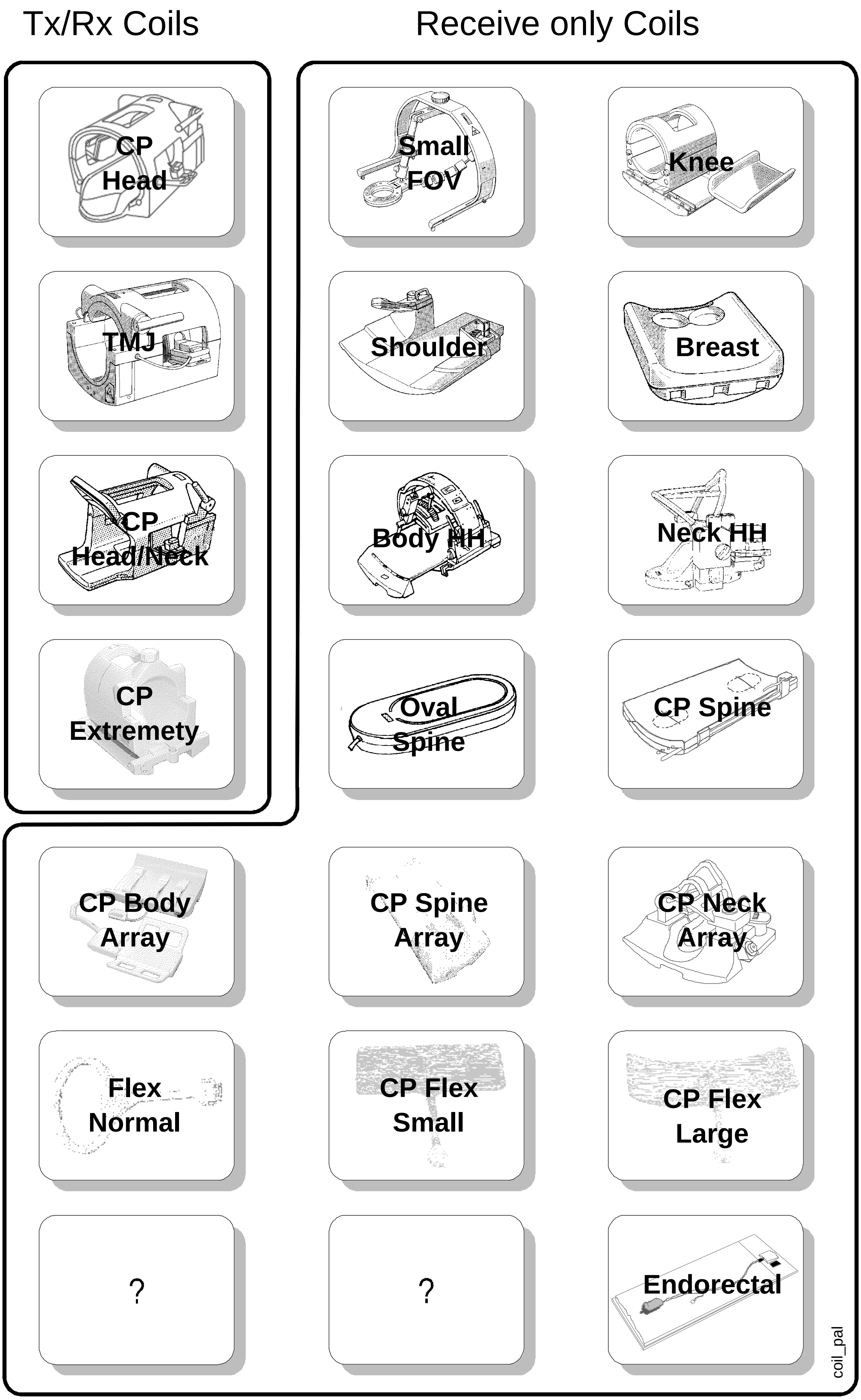
Tuning

The CP Body coil and CP Head/Neck coil have motorized tuning capacitors for impedance matching. The tuning procedure involves setting the capacitor values to match the coil impedance to the RF line impedance for maximum power transfer. With help from the newly developed tuning method, the "Exact Tuning" algorithm (ET/a)", it is possible after an initial tune measurement to calculate the motor/capacitor positions required to tune the coils. This assures a very quick and accurate tuning. The **MOT_CON** board supplies here the necessary motor drive control and regulation.

The tunable local coils utilize varicap diodes as tuning elements. The **VAR_CON** provides the essential linear DC voltages to the varicap diodes.

DCU

The Display Control Unit with its plasma display and function keys, serves as the operators control for the operation of table movements, coil tuning, coil identification and PMU functions.


Tx/Rx Coils
Receive only Coils
CP Head
Small FOV
Knee
TMJ
Shoulder
Breast
CP Head/Neck
Body HH
Neck HH
CP Extremety
Oval Spine
CP Spine
CP Body Array
CP Spine Array
CP Neck Array
Flex Normal
CP Flex Small
CP Flex Large
?
?
Endorectal
coil_pal

Antenna Systems

Image quality can only be as good as the receiving components allow and the antennas of course become a major factor. Antenna designs need to accommodate the peculiar formation of somatic extremities in order to optimize the RF field homogeneity and pick-up gain. The locations and dimensions of these regions will dictate localized volume and homogeneity requirements. A whole family of antennas have been designed to meet the specialized needs of diagnostic imaging with MR.

Body Coil

A circular polarized (CP) Body coil is used for whole-body imaging and for transmission in conjunction with the specialized local coil antennas.

The Body coil has been considerably improved over previous CP designs. The new design takes patient dimensions and it's effect on cross-talk into account. An **E-field blocking system** has been developed and employed to avoid excessive coil loading in the shoulder region. Improved and optimized **coil dimensions** have brought about an increase in the signal-to-noise ratio as well.

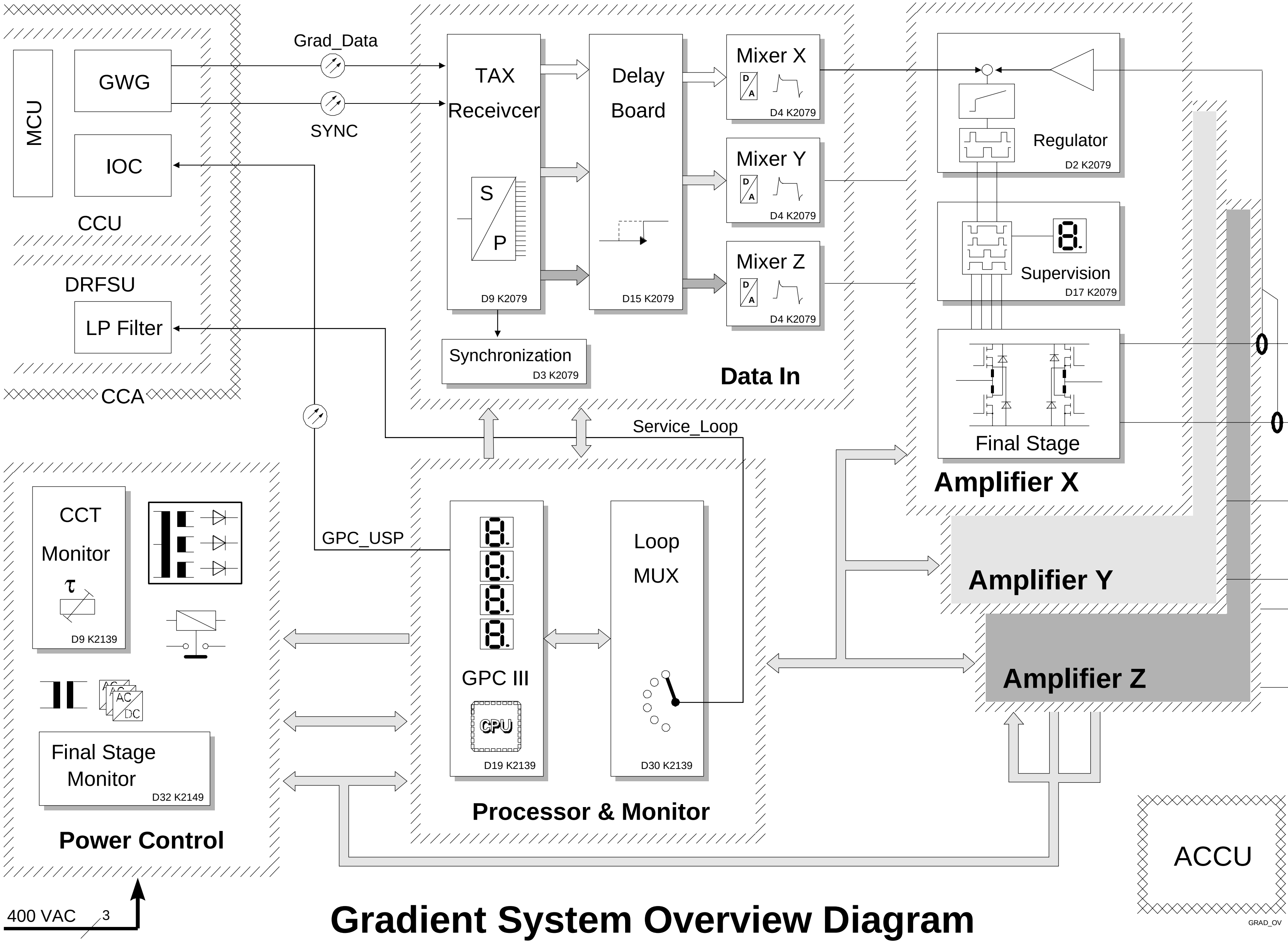
Additionally, it has been made extremely **service friendly**. It consists of only six components: four resonator elements and two tuning boxes. In the unlikely event any one of these components were to become defective, quick-disconnect type connectors promote **quick and easy component replacement**.

Local Coils

To accommodate the high demand for high-resolution antenna types, Siemens has produced a very wide range of anatomical imaging coils. For reasons already mentioned, three coils have been designed, varying in price and performance, for the head and head/neck regions as well as three coil types for partial of full spinal coverage. Other coil types exist designed especially for or suited to image practically every part of the human body.

Array Coil

Trend setting developments in the Array coil arena have lead to the production of antenna assemblies which can cover a very wide area and at the same time provide excellent signal quality. An array coil consists of several resonator elements in an array configuration which can be variably selected. This gives a wide range of coverage while maintaining the active coil size to within confines giving it an optimal S/N figure.



Gradient System Overview Diagram

Gradient System

The SIEMENS gradient system is currently the best developed and performing gradient amplifier and coil system the industry has to offer. It's efficient and effective design make it a first-class performer, extremely reliable, and very service friendly.

The gradient amplifier employs a switch-mode design. It is capable of producing enough power to satisfy even the most gradient hungry sequences developed to date while achieving a rate of performance unexcelled anywhere in the MR industry today.

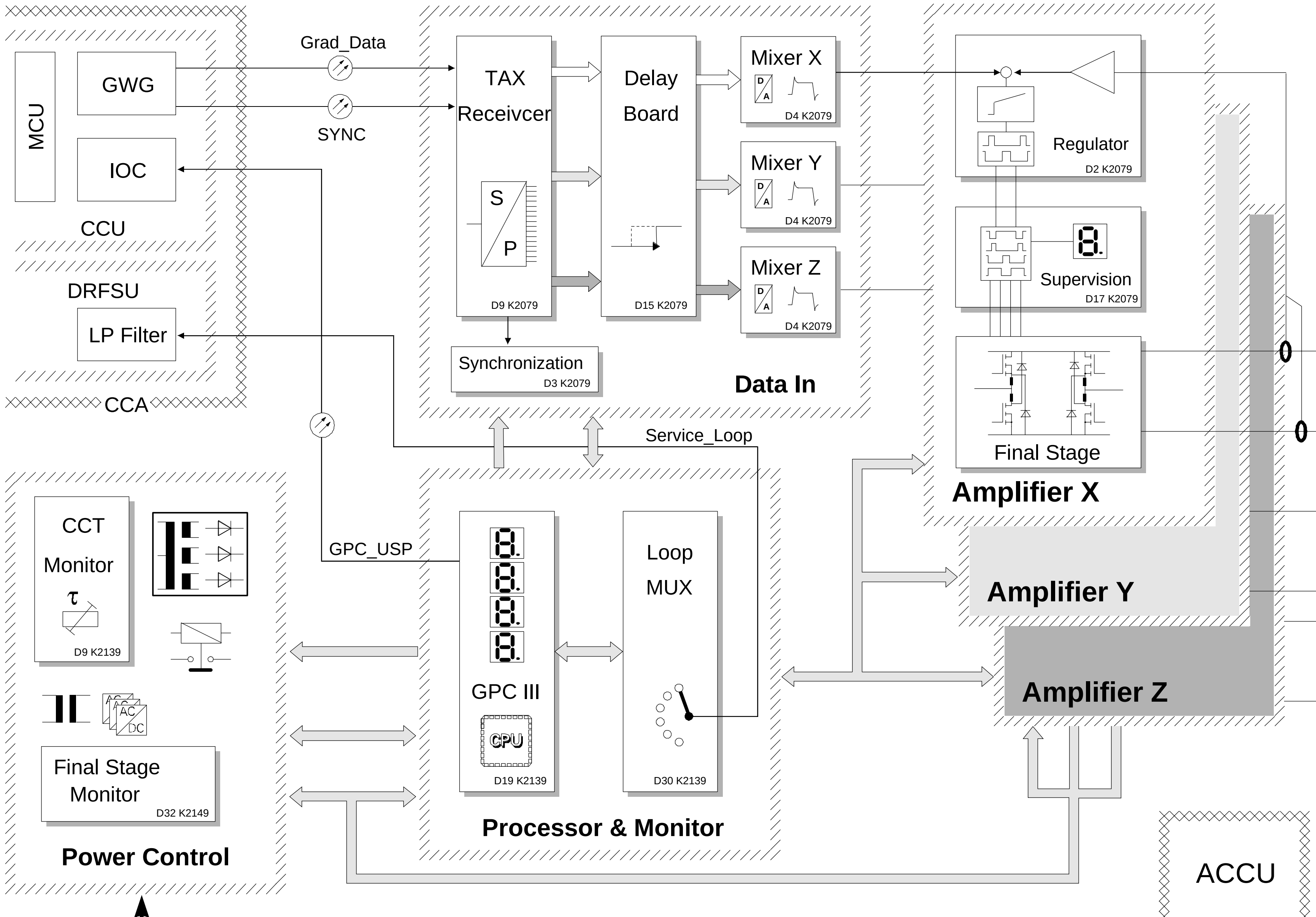
The amplifier contains monitoring circuitry which covers all majors facets of amplifier operation. Status display and indicators provide the service engineer with a host of error messages which help direct the service engineer quickly to the probable cause of trouble. It's inherent redundancy in hardware components (there are three axis') allow on-the-site part swapping for many of the components when performing trouble shooting routines giving yet another edge in service strategy. Here are a few excerpts from the specifications of the units capabilities :

25 mT/m AS Gradient System

- Gradient strength : max. 25mT/m
- Rise time 0 to 25 mT/m : 0.6 ms
- Final Stage supply voltage : 600 Volt
- max. current (for 25 mT/m) : 250 Ampere
- Duty cycle : 100%
- Water cooling for final stages : max. 30l/min.

25 mT Gradient Coil

- Actively shielded coil system for all directions
  (active eddy current compensation)
- Water cooling for coil system max. 20l/min.
- Weight 1700 kg



Gradient System Overview Diagram

To gain an overview of the amplifier, we can divide the components into 4 groups :

- Power Control
- Data In
- Amplifier
- Processor & Monitor

Power Control

The Power Control section consists of the control electronic power supplies (+5, +15, -15) and 600V supply for the final stages in the Amplifier section. It also provides the power up (on/off) control and monitoring. The final stage supply voltage of 600V allow ramp times of 25mT/m within 600µs. The amplifier monitoring section keeps an eye on the final stage supply voltage.

Data In

The Data In section is responsible for receiving the digital gradient wave forms from the GWG, delaying and converting them to an analog signal and finally pre-distort them for the eddy current compensation. The D/A converters of the Mixer boards are offer 18 bit resolution providing a high degree of stability and accuracy.

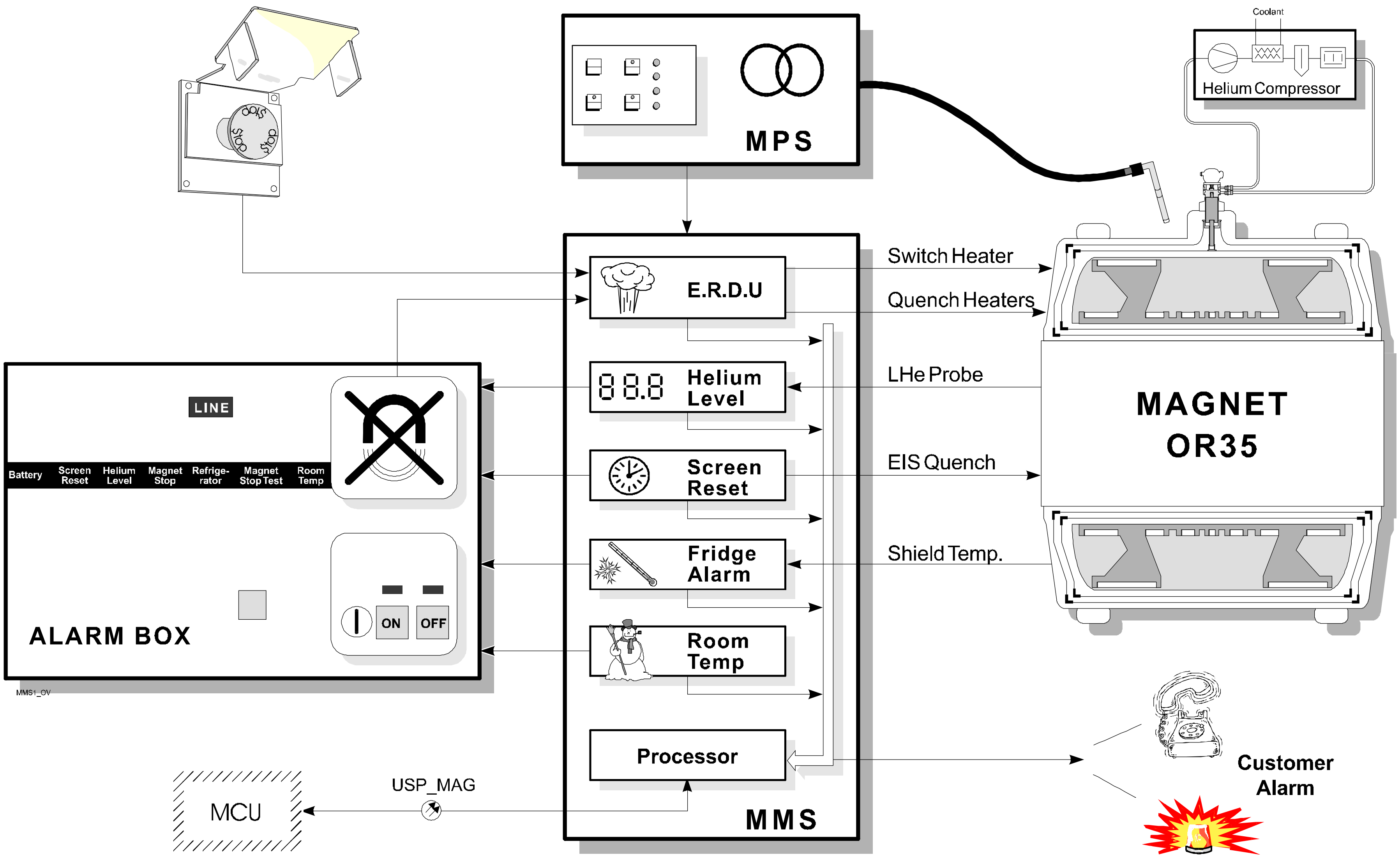
Amplifier

The amplifier consists of the regulator section and final stages. The nominal value is given by the Data In section and the amplifier produces the current pulses to the gradient coil. The output current is produced by switching 600V to the coil over a FET bridge which is driven by PWM control signals from the regulator. The duty cycle of the PWM signals determines the amount of current, and direction of the current is controlled by selecting the proper FET pair of the bridge.

Processor and Monitoring

The Processor and Monitoring has the task of enabling the amplifier and monitoring the vital statistics of the amplifier stages. The monitoring assures the hardware is protected from serious damage.

Magnet System Overview Diagram



Magnet System

One of the main prerequisites for MRI systems is a strong and highly homogeneous magnetic field inside the measurement volume. The homogeneous magnetic field is produced, maintained, and monitored by the following components:

- **MPS:** Magnet Power Supply
- **MMS:** Monitoring system for magnet and refrigerator. The MMS is also responsible for system communication.
- **Alarm Box**: The alarm box generates visual and / or audible error indications. The box also contains the quench release button and the ON/OFF key switch for main power control of the system
- **OR 35 Magnet:** Superconducting magnet with refrigerator

Magnet Power Supply (MPS)

The MPS contains the components for ramping the OR 35 and includes the power and control electronics, switch heater supply and safety electronics. It is capable of delivering up to 725 A for ramping up the magnet to a field strength of 1.5 T.

Magnet Monitoring System (MMS)

The MMS comprises:

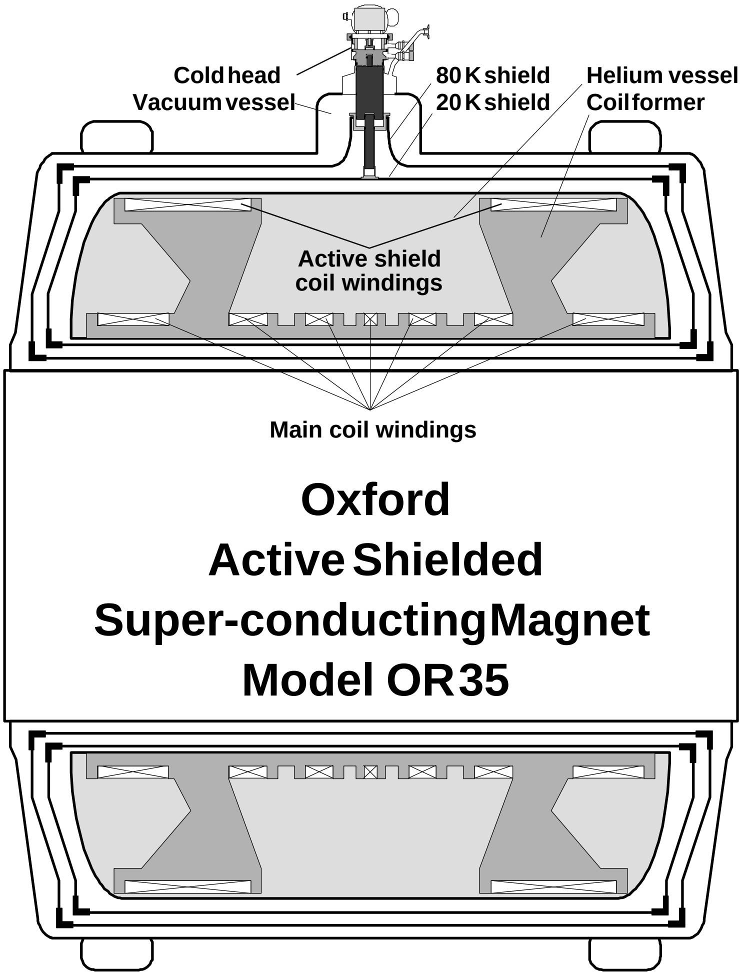
The MMS comprises:

- **ERDU** (**E**mergency **R**un **D**own **U**nit). This unit quenches the magnet in case of emergency.
- The **LHe LEVEL METER** samples in time intervals and continuously indicates the helium fill level. The display is located in the upper part of the PCA.

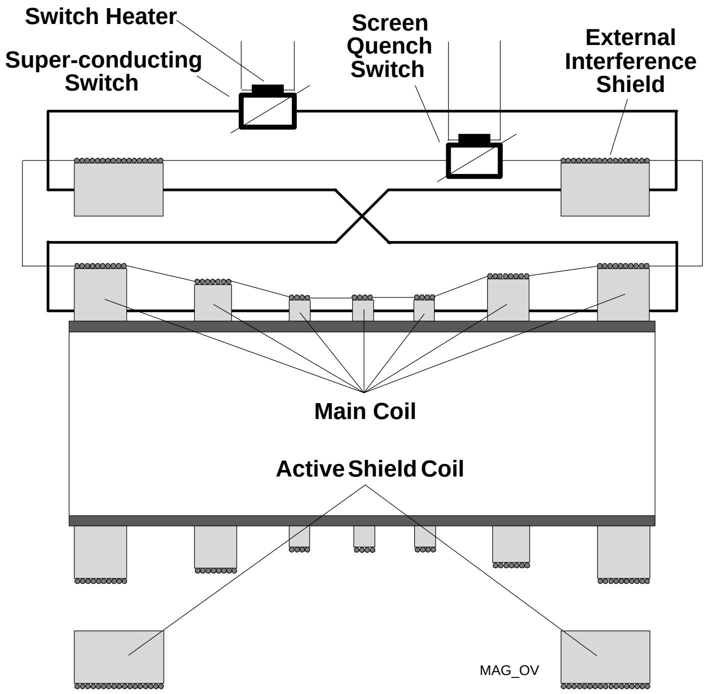
The **FRIDGE ALARM** circuit monitors the shield temperature and a LED indicates the status of the refrigerator.

Alarm Box

The Alarm Box indicates the status of the magnet and the refrigerator. Errors detected by the MMS will be visually and audibly indicated on the Alarm Box. In addition, the quench release button and the key main power ON/OFF switch of the system is included on the alarm box, which is located in the operator room next to the operating console.



Cold head
Vacuum vessel
80 K shield
20 K shield
Helium vessel
Coil former
Active shield
coil windings
Main coil windings
Oxford
Active Shielded
Super-conducting Magnet
Model OR 35



Switch Heater
Super-conducting
Switch
Screen
Quench
Switch
External
Interference
Shield
Main Coil
Active Shield Coil
MAG_OV

OR 35 Magnet

The OR 35 is a superconducting **AS** (**A**ctive **S**hielded) magnet produced by our subsidiary OMT and has a nominal field of 1.5T. To reduce helium boil-off losses, a compressor and UNISOCK coldhead system are implemented to cool the 20k and 80k (Kelvin) shields.

B_0-Screen

The B_0-screen or **EIS** (**E**xternal **I**nterference **S**creen) is located on each superconducting coil section of the magnet. Because these screen windings are shorted together, any transient interference will be absorbed by the screen, preventing any disturbance of the main field.

Active shielding

The windings are arranged so that the external field is compressed (relative to "normal" magnets), minimizing the external zone of influence of the magnet. This configuration known as "active shielding" is a patent of Oxford Magnet Technology.

Cryostat

The outer vacuum chamber (OVC) contains all the magnet hardware and the helium vessel that is placed in a vacuum. Suspensions of low thermal conductivity support the internal components to hold them mechanically in place without building "thermal conductive bridges" to the outside. Inside the insulation, solid aluminum shells (shields) are maintained at cryogenic temperature by an externally mounted refrigeration unit (coldhead). Heat which reaches this level is absorbed and passed to an external heat exchanger via the refrigerator.

UNISOCK

The refrigerator unit (coldhead) is mounted on a UNISOCK (patent pending) which facilitates positioning and replacing the refrigerator head.

The MPS and MMS are installed in the PCA (**P**ower **CA**binet) K2126.

Abbreviations

Cabinets

ACCU	Air Compressor Chiller Unit
CCA	Control CAbinet
EPI	Êcho Planar Imaging (Booster
GPS	Gradient Power Supply
PCA	Power CAbinet
RFPA	Radio Frequency Power Amplifier

Assemblies

CCU	Control and Communication Unit
DCU	Display Control Unit
DDS	Data Documentation System
DRFSU	Digital Radio Frequency Signal Unit
GPA	Gradient Power Amplifier
LPD	Line Power Distribution
MMS	Magnet Monitoring System
MPS	Magnet Power Supply
MRC	MR Console
MRSC	MR Satellite Console
PT	Patient Table
RFAS	RF Application System
RFCC	Radio Frequency Coil Control
RFPA	Radio Frequency Power Amplifier
SCU	Sequence Control Unit (SEQ + RFWG + GWG+ARDAS-CON+PMU2)
SMI	Siemens Medical Imager
SHIM	.

PC Boards and Components

(listed according to the assemblies)

HOST:	SPARC 10	.
	CIB	Cabinet Interface Board
	KII	Keyboard and Intercom Interface
SMI-5:	SIAM-C	System Interface And Memory - with Cable
	IMAGER-ML	.
	MAP 4	Memory And Processors (4)
	DATA-I/O	.
	MAXICAM	.
CCU	ARDAS-CON	ARray Data Acquisition System CONtroller
	GWG	Gradient Waveform Generator
	IOC	Input Output Controller
	MCU	Main Control Unit
	PMU2	Physiological Measurement Unit 2
	PMU_DAS	PMU Data Acquisition System
	RFWG	Radio Frequency Waveform Generator
	SEQ	Sequencer
DRFSU	ISO-B	ISOlation Board (of type) B
	PALI X	Power Absorption Limit Indicator X
	SYSCO	System Control
	TRACODI	TRansmitter Analog COntrol Digital
	TUSUPI	TUne and SUPervision Interface
RFAS	AMCOMUX	Amplifier, Coupler, Multiplexer (MUX)
	HYBRID	(T/R switch for CP coils)
	PS	Power Splitter (= Hybrid)
	T/R	Transmit / Receive switch
RFCC	MOT_CON	MOTor CONtroller
	PIN_CON	PIN diode CONtrol
	PNEU_CON	PNEUmatic switch CONtrol
	TABLE_CON	patient TABLE CONtrol
	VAR_CON	VARicap tuning diode CONtrol
GPA	GPC	Gradient Processor Controller
MMS	ERDU	Emergency Run Down Unit (Magnet Stop)
MPS		

General

ABR	ABsorption Rate (the rate and amount of RF transmit into patient)
ADC	Analog to Digital Converter
AS	Active Shielded
ASIC	Application Specific Integrated Circuit
B	Body (coil) - used in many signal names
B	mathm. quantity of a magnetic field
CC	Constant Current (source)
C	Circular Polarization
CPU	Central Processing Unit
DRAM	Dynamic Random Access Memory
DYNCON	DYNamic CONtrol
E	mathm. quantity of an electric field
EIS	External Interference Screen
ET/a	Exact Tuning Algorithm
FFT	Fast Fouier Transform
FOV	Field OF View
G-Bus	Graphics Bus
GIPSY	General Interface Processor for SYstem-bus
HTE	Hardware Test Environment
IC	Integrated Circuit
I/O	Input Output
ISD	Integrated Services Digital Network
LC	Local Coil - also used in signal names
LHE, LHe	Liquid Helium
LISA	Linear Interpolation and Shrink ASIC (zooming)
LP	Linear Polarization, Low Pass (filter)
M-Bus	Memory Bus
MAGIC	Multimux Address Generator and Interface Controller
MMX	Multiple MultipleX
MOD	Magnetic Optical Disk (drive)
MRI	Magnetic Resonance Imaging
O/S	Output Short (contactor)
OVC	Outer Vacuum Chamber
PCB, PC	Printed Circuit (Board)
PEP	Peak Envelope Power

PIN	P-Intrinsic-N junction diodE
PSU	Power Supply Unit
QH	Quench Heater
RF	Radio Frequency
ROI	Region Of Interest
Rx	Receive
S-Bus	SUN Bus
SCON	S-Bus CONnector
SCS	Superconducting Switch
SCSI	Small Computer System Interface
SEC	
SH	Switch Heater
S/N	Signal-to-Noise
SPCI	Standard Protocol of Camera Interface
SPDI	Siemens Parallel Digital Interface
SSB	Single Side-Band
T	Tesla - mathm. unit of magnetism
TICO	Timer and Interrupt COntroller
T/R	Transmit/Receive
Tx	Transmit
USP	Universal Serial (communication) Protocol

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All Chapters:

Editorial changes

New layout

Chapter 1

Pg 4 and 5	Blockdiagram of options added

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